# OXIDATION AND REDUCTION

3

**Topics covered in this chapter:**

## 3.1 OXIDATION AND REDUCTION

The early definition of an *oxidation* reaction was simply the combining of a substance with oxygen. Magnesium, for example, combines with oxygen to form MgO while carbon will burn to form $CO_2$.

$$2Mg_{(s)} + O_{2(g)} \rightarrow 2MgO_{(s)}$$

$$2C_{(s)} + O_{2(g)} \rightarrow 2CO_{(s)}$$

Similarly, *reduction* was considered as the opposite to oxidation and occurred when a substance lost oxygen. A typical example being the reduction of iron ore to iron metal.

$$Fe_2O_{3(s)} + 3CO_{(g)} \rightarrow 2Fe_{(l)} + 3CO_{2(g)}$$

It also became evident that oxidation and reduction reactions are interdependent and always occur together. Hence the term redox reaction (*red*uction - *ox*idation).

oxidation

$$Mg_{(s)} + H_2O_{(g)} \rightarrow MgO_{(s)} + H_{2(g)}$$

reduction

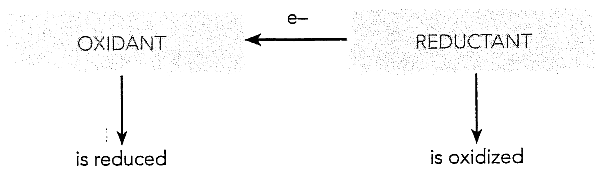
### Redox reactions – a more general view

The early definitions of redox reactions are far too limiting and do not cover many similar reactions which don't involve oxygen. Redox reactions, in fact, simply involve the transfer of electrons from one species to another.

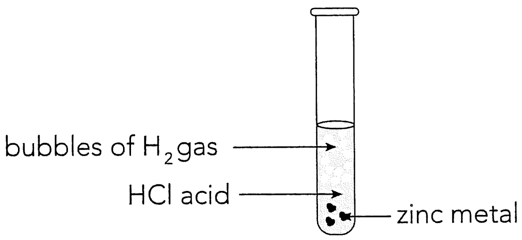
Oxidation - the loss of electrons.

Reduction - the gain of electrons.

Two quite different examples are as follows:




e.g. 1. The reaction between zinc and hydrochloric acid.




reduction

$$Zn_{(s)} + 2H^+_{(aq)} \xrightarrow{} Zn^{2+}_{(aq)} + H_{2(g)}$$

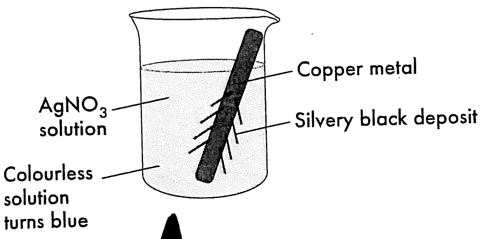
2e-

oxidation

$Zn_{(s)} \rightarrow$ loses electrons ∴ is oxidised.

$H^+_{(aq)} \rightarrow$ gains electrons ∴ is reduced.

e.g. 2. The reaction between copper metal and silver nitrate solution.



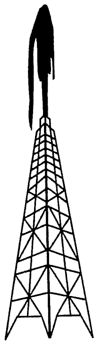

$2e^-$ oxidation

$$2Ag^+_{(aq)} + Cu_{(s)} \rightarrow 2Ag_{(s)} + Cu^{2+}_{(aq)}$$

reduction

$Cu_{(s)} \rightarrow$ loses electrons $\therefore$ is oxidised.

$Ag^+_{(aq)} \rightarrow$ gains electrons $\therefore$ is reduced.



OIL Oxidation Is Loss of electrons

RIG Reduction Is Gain of electrons

## Oxidising and reducing agents

Oxidation and reduction occur at the same time in a redox reaction. A redox reaction involves the transfer of electrons from the substance being oxidised to that being reduced.

*Oxidising agents* or *oxidants* cause a substance to be oxidised. They have a tendency to accept electrons and are themselves reduced.

e.g. $O_2$ $Cl_2$ $MnO_4^-$ $Cr_2O_7^{2-}$ $ClO^-$ $H^+$ conc. $H_2SO_4$ conc. $HNO_3$

*Reducing agents* or *reductants* cause a substance to be reduced. They have a tendency to donate electrons and are themselves oxidised.

e.g. Zn C $H_2$ $Fe^{2+}$ $C_2O_4^{2-}$

$2e^-$ oxidation

e.g. $PbO_{(s)} + CO_{(g)} \rightarrow Pb_{(s)} + CO_{2(g)}$

reduction

**PbO**
- accepts electrons
- is reduced to Pb
- is the oxidant

**CO**
- donates electrons
- is oxidised to $CO_2$
- is the reductant

### Question 3.1

The burning of magnesium metal in chlorine gas results in the formation of $MgCl_2(s)$.

(a) Write the equation for this reaction.

________________________________________

(b) Which species loses electrons? ____________________

(c) Which species is oxidised? ____________________

(d) Which species is the oxidant? ____________________

Question 3.2

Redox reactions involve competition for electrons. Is this statement true or false? Use examples to support your reasoning.

## Oxidation numbers

Oxidation numbers are a convenient way of determining if a substance has been oxidised or reduced. These numbers are arbitrarily assigned to atoms and are equal to the charge the atom would have if its bonds were purely ionic.

| RULES FOR ASSIGNING OXIDATION NUMBERS | |
|---|---|
| **Species** | **Oxidation Number** |
| 1. Atoms in the elemental state | 0 |
| 2. Monatomic ions<br>(Group I metals in combined state)<br>(Group II metals in combined state) | charge on the ion<br>+1<br>+2 |
| 3. Oxygen in combined state<br>Exception 1 : peroxides e.g. $Na_2O_2$ , $H_2O_2$<br>Exception 2 : $F_2O$ | –2<br>–1<br>+2 |
| 4. Hydrogen in combined state<br>Exception : metal hydrides e.g. NaH | +1<br>–1 |
| 5. For polyatomic species the sum of the oxidation numbers | charge on the ion |

## Worked Example

3.1 Determine the oxidation number of Mn in $Mn_2O_3$ and $MnO_4^-$.

(a) $Mn_2O_3$ :

$$
\begin{aligned}
2 \times (Mn) + 3 \times (O) &= 0 \\
2 \times (Mn) + 3 \times (-2) &= 0 \\
2 \times (Mn) &= +6 \\
Mn &= +3
\end{aligned}
$$

(b) $MnO_4^-$ :

$$
\begin{aligned}
(Mn) + 4 \times (O) &= -1 \\
(Mn) + 4 \times (-2) &= -1 \\
Mn - 8 &= -1 \\
Mn &= +7
\end{aligned}
$$

Question 3.3

Determine the oxidation numbers of each element in the following:

(a) $Cr_2O_7^{2-}$ : Cr ________ O ________

(b) $CrO_4^{2-}$ : Cr ________ O ________

(c) $ClO_3^-$ : Cl ________ O ________

(d) $NH_4Cl$ : N ________ H ________ Cl ________

(e) $HSO_3^-$ : H ________ S ________ O ________

(f) $CaCO_3$ : Ca ________ C ________ O ________

## 3.2 IDENTIFYING OXIDATION – REDUCTION REACTIONS

By assigning oxidation numbers to all the atoms involved in a reaction, we can determine if oxidation or reduction has occurred.

- oxidation – an increase in oxidation number
- reduction – a decrease in oxidation number

e.g.

$$\underset{0}{Cu_{(s)}} + \underset{+1}{4H^+_{(aq)}} + \underset{+5\ -2}{2NO_3^-{}_{(aq)}} \rightarrow \underset{+4\ -2}{2NO_{2(g)}} + \underset{+2}{Cu^{2+}_{(aq)}} + \underset{+1\ -2}{2H_2O_{(l)}}$$

oxidation numbers

reduction(-1)

oxidation(+2)

- Cu has been oxidised (O.N. 0 → +2)
- N has been reduced (O.N. 5 → 4)
- All other species have retained the same oxidation number, hence they have
- neither been reduced nor oxidised.
- This reaction is a redox reaction (both oxidation and reduction have taken place).

Question 3.4

Assign oxidation numbers to the following to determine if they are redox reactions.

(a) $Sn^{2+}{}_{(aq)} + Cl_{2(g)} \rightarrow 2Cl^-{}_{(aq)} + Sn^{4+}{}_{(aq)}$

(b) $H_2CO_{3(aq)} \rightarrow H_2O_{(l)} + CO_{2(g)}$

(c) $2CrO_4^{2-}{}_{(aq)} + 2H^+{}_{(aq)} \rightarrow Cr_2O_7^{2-} + H_2O_{(l)}$

(d) $Ca_{(s)} + 2H_2O_{(l)} \rightarrow Ca(OH)_{2(s)} + H_{2(g)}$

## Half-equations

It is often convenient to consider a redox reaction by writing two half equations. One equation represents oxidation while the other represents reduction.

- oxidation half equation $Zn_{(s)} \rightarrow Zn^{2+}{}_{(aq)} + 2e^-$
- reduction half equation $Cu^{2+}{}_{(aq)} + 2e^- \rightarrow Cu_{(s)}$

- redox equation $Zn_{(s)} + Cu^{2+}{}_{(aq)} \rightarrow Zn^{2+}{}_{(aq)} + Cu_{(s)}$

### Question 3.5

Write half-equations for:

(a) $Zn_{(s)} + 2H^+{}_{(aq)} \rightarrow Zn^{2+}{}_{(aq)} + H_{2(g)}$

oxidation ______________________________

reduction ______________________________

(b) $Mg_{(s)} + S_{(s)} \rightarrow MgS_{(s)}$

oxidation ______________________________

reduction ______________________________

## Balancing Half-equations

1. Use oxidation numbers to identify the element being oxidised or reduced.
2. Write a skeleton equation (reactant → product).
3. Balance the number of atoms of the element being oxidised or reduced.
4. Balance **oxygen** atoms by adding $H_2O$.
5. Balance **hydrogen** atoms by adding $H^+$.
6. Balance **charge** by adding electrons to the side that is most positive.

## Worked Example

3.2 Write a half equation for the oxidation of:
(a) $NO_3^-{}_{(aq)}$ to $NO_{(g)}$ in acidic solution.
(b) $PbO_{2(s)}$ to $Pb^{2+}{}_{(aq)}$ in acidic conditions.

(a) $NO_3^-{}_{(aq)} \rightarrow NO_{(g)}$ *skeleton equation containing element reduced (N).*

$NO_3^-{}_{(aq)} \rightarrow NO_{(g)} + 2H_2O_{(l)}$ *oxygen balanced*

$NO_3^-{}_{(aq)} + 4H^+{}_{(aq)} \rightarrow NO_{(g)} + 2H_2O_{(l)}$ *hydrogen balanced*

$NO_3^-{}_{(aq)} + 4H^+{}_{(aq)} + 3e^- \rightarrow NO_{(g)} + 2H_2O_{(l)}$ *charges balanced*

It is not necessary to actually write out each step – the previous page is an illustration of the steps involved.

(b) Similiary to (a).

$$PbO_{2(s)} \rightarrow Pb^{2+}{}_{(aq)}$$

$$PbO_{2(s)} \rightarrow Pb^{2+}{}_{(aq)} + 2H_2O_{(l)}$$

$$PbO_{2(s)} + 4H^+{}_{(aq)} \rightarrow Pb^{2+}{}_{(aq)} + 2H_2O_{(l)}$$

$$PbO_{2(s)} + 4H^+{}_{(aq)} + 2e^- \rightarrow Pb^{2+}{}_{(aq)} + 2H_2O_{(l)}$$

### Question 3.6

Complete the following half-equations for acidic conditions.

(a) $MnO_4{}^-{}_{(aq)} \rightarrow Mn^{2+}{}_{(aq)}$

(b) $H_2C_2O_{4(aq)} \rightarrow CO_{2(g)}$

(c) $CH_3CH_2OH_{(l)} \rightarrow CH_3CHO_{(l)}$

## 3.3 BALANCING REDOX EQUATIONS – USING HALF-EQUATIONS

To obtain an overall reaction it is best to consider and balance each half equation first. These can then be added so that the number of electrons for each half equations is equal.

### Worked Example

3.3 Balance the following redox reaction.

$$Cu_{(s)} + NO_3{}^-{}_{(aq)} + H^+{}_{(aq)} \rightarrow Cu^{2+}{}_{(aq)} + NO_{2(g)} + H_2O_{(l)}$$

Consider the principal reactants and write balanced half-equations.

Reduction

$$NO_3{}^-{}_{(aq)} \rightarrow NO_{2(g)}$$

$$NO_3{}^-{}_{(aq)} \rightarrow NO_{2(g)} + H_2O_{(l)}$$

$$NO_3{}^-{}_{(aq)} + 2H^+{}_{(aq)} + e^- \rightarrow NO_{2(g)} + H_2O_{(l)}$$

Oxidation

$$Cu_{(s)} \rightarrow Cu^{2+}{}_{(aq)} + 2e^-$$

Add the two half equations so that electrons cancel.

$$(NO_3{}^-{}_{(aq)} + 2H^+{}_{(aq)} + e^- \rightarrow NO_{2(g)} + H_2O_{(l)}) \quad \times 2$$

$$(\qquad Cu_{(s)} \rightarrow Cu^{2+}{}_{(aq)} + 2e^-) \quad \times 1$$

Redox equation

$$Cu_{(s)} + 2NO_3{}^-{}_{(aq)} + 4H^+{}_{(aq)} \rightarrow Cu^{2+}{}_{(aq)} + 2NO_{(g)} + 2H_2O_{(l)})$$

## Question 3.7

Balance the following using half-equations.

(a) $MnO_4^-{}_{(aq)} + H_2S_{(g)} + H^+{}_{(aq)} \rightarrow Mn^{2+}{}_{(aq)} + S_{(s)} + H_2O_{(l)}$

(b) $ClO^-{}_{(aq)} + SO_{2(g)} + H_2O_{(l)} \rightarrow Cl^-{}_{(aq)} + SO_4^{2-}{}_{(aq)} + H^+{}_{(aq)}$

## Question 3.8

Balance the following equations.

(a) $Fe^{2+}{}_{(aq)} + MnO_4^-{}_{(aq)} + H^+{}_{(aq)} \rightarrow Fe^{3+}{}_{(aq)} + Mn^{2+}{}_{(aq)} + H_2O_{(l)}$

(b) $Sn^{2+}{}_{(aq)} + H^+{}_{(aq)} + OCl^-{}_{(aq)} \rightarrow Cl^-{}_{(aq)} + Sn^{4+}{}_{(aq)} + H_2O_{(l)}$

## Disproportionation reactions*

Under certain conditions some substances are able to undergo self oxidation and reduction. These types of reactions are called disproportionation reactions. Two important examples are:

(i) The decomposition of hydrogen peroxide.

reduction (-1)

$$H_2O_{2(aq)} + H_2O_{2(aq)} \rightarrow 2H_2O_{(l)} + O_{2(g)}$$

oxidation numbers → 1 -1    +1 -1    1 -2    0

oxidation (+1)

or simply $$2H_2O_{2(aq)} \rightarrow 2H_2O_{(l)} + O_{2(g)}$$

(ii) The disproportionation of copper (I) ions in aqueous solution.

oxidation (+1)

$$Cu^+_{(aq)} + Cu^+_{(aq)} \rightarrow Cu^{2+}_{(aq)} + Cu_{(s)}$$

reduction (-1)

or simply $$2Cu^+_{(aq)} \rightarrow Cu^{2+}_{(aq)} + Cu_{(s)}$$

### Question 3.9

In acidic conditions the chlorate ion ($ClO_3^-$) undergoes disproportionation to $Cl_{2(g)}$ and $ClO_4^-{}_{(aq)}$. Write appropriate half-equations and the overall equation.

### Question 3.10

Sodium hypochlorite, NaOCl, can be prepared by bubbling chlorine gas into a solution of NaOH. Assuming the $Cl^-$ ion is also produced, write appropriate half equations to show the disproportionation of $Cl_2$.

* *May not be required for your course.*

## 3.4 COMPETITION FOR ELECTRONS

Redox reactions involve a competition for electrons. The species being reduced has a greater ability or potential to gain electrons than the substance being oxidised. It is possible to set up experimental apparatus that can be used to compare chemical species potentials to be reduced or oxidised. By convention, the reduction of $H^+$ to $H_2$ is chosen as the standard reduction reaction to compare the potential of all other species. As it is the standard, the reduction of $H^+$ is said to have a potential of 0.00V

i.e. $2H^+_{(aq)} + 2e^- \rightarrow H_{2(g)}$ $E° = 0.00$ V (1.0 mol $L^{-1}$, 25°C)

It is important to remember that reduction potentials cannot be measured separately; they must be measured with a species that is simultaneously being oxidised.

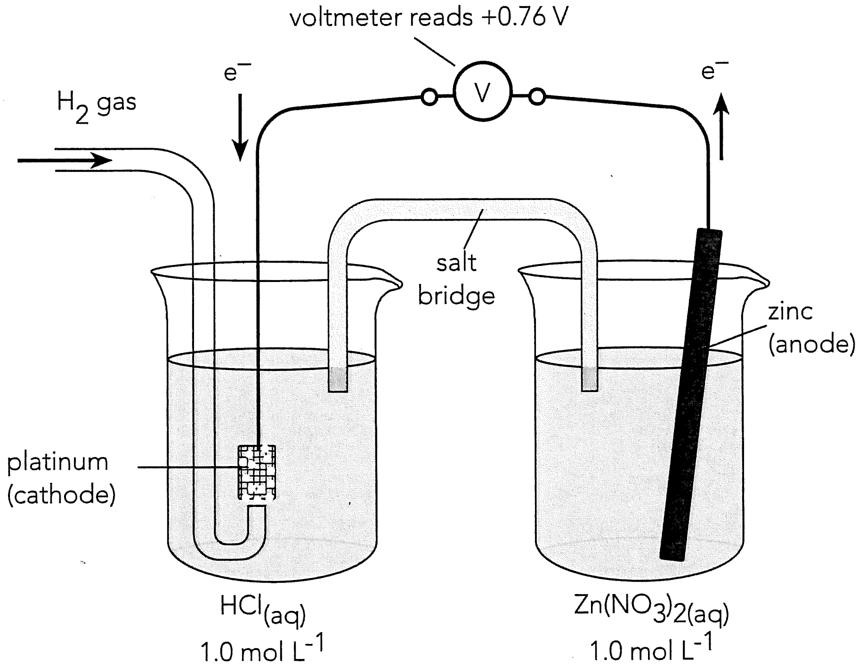



*Figure 3.1 Measuring standard reduction potential of a Zn/$Zn^{2+}$ half-cell. The $H^+$/$H_2$ half-cell is used as a reference cell. The salt bridge is a glass tube filled with electrolyte solution. This allows ion migration (transfer or charge).*

### Standard reduction potentials (E° values)

- are measured using the hydrogen half-cell as a reference cell
- apply to 1.0 mol $L^{-1}$ solutions, 101.3 kPa gas pressure and 25°C temperature
- are a measure of the relative tendency of a species to be reduced
- can be used to predict:
  - electrochemical cell voltages
  - whether particular redox reactions could occur.

## Some Standard Reduction Potentials

Strong oxidising agents

Weak reducing agents

| Half-reaction | | | E° (volts) |
|---|---|---|---|
| $F_2(g) + 2e^-$ | → | $2F^-(aq)$ | +2.89 |
| $H_2O_2(aq) + 2H^+(aq) + 2e^-$ | → | $2H_2O(l)$ | +1.76 |
| $PbO_2(s) + SO_4^{2-}(aq) + 4H^+(aq) + 2e^-$ | → | $PbSO_4(s) + 2H_2O(l)$ | +1.69 |
| $MnO_4^-(aq) + 8H^+(aq) + 5e^-$ | → | $Mn^{2+}(aq) + 4H_2O(l)$ | +1.51 |
| $Au^{3+}(aq) + 3e^-$ | → | $Au(s)$ | +1.50 |
| $Cl_2(g) + 2e^-$ | → | $2Cl^-(aq)$ | +1.36 |
| $Cr_2O_7^{2-}(aq) + 14H^+(aq) + 6e^-$ | → | $2Cr^{3+}(aq) + 7H_2O(l)$ | +1.36 |
| $O_2(g) + 4H^+(aq) + 4e^-$ | → | $2H_2O(l)$ | +1.23 |
| $Br_2(l) + 2e^-$ | → | $2Br_2^-(aq)$ | +1.08 |
| $NO_3^-(aq) + 4H^+(aq) + 3e^-$ | → | $NO(g) + 2H_2O(l)$ | +0.96 |
| $ClO^-(aq) + H_2O(l) + 2e^-$ | → | $Cl^-(aq) + 2OH^-(aq)$ | +0.90 |
| $Ag^+(aq) + e^-$ | → | $Ag(s)$ | +0.80 |
| $Fe^{3+}(aq) + e^-$ | → | $Fe^{2+}(aq)$ | +0.77 |
| $O_2(g) + 2H^+(aq) + 2e^-$ | → | $H_2O_2(aq)$ | +0.70 |
| $I_2(s) + 2e^-$ | → | $2I^-(aq)$ | +0.54 |
| $Cu^{2+}(aq) + 2e^-$ | → | $Cu(s)$ | +0.34 |
| $2H^+(aq) + 2e^-$ | → | $H_2(g)$ | 0 (exactly) |
| $Pb^{2+}(aq) + 2e^-$ | → | $Pb(s)$ | -0.13 |
| $Sn^{2+}(aq) + 2e^-$ | → | $Sn(s)$ | -0.14 |
| $Ni^{2+}(aq) + 2e^-$ | → | $Ni(s)$ | -0.24 |
| $Fe^{2+}(aq) + 2e^-$ | → | $Fe(s)$ | -0.44 |
| $Zn^{2+}(aq) + 2e^-$ | → | $Zn(s)$ | -0.76 |
| $2H_2O(l) + 2e^-$ | → | $H_2(g) + 2OH^-(aq)$ | -0.83 |
| $Al^{3+}(aq) + 3e^-$ | → | $Al(s)$ | -1.68 |
| $Mg^{2+}(aq) + 2e^-$ | → | $Mg(s)$ | -2.36 |
| $Na^+(aq) + e^-$ | → | $Na(s)$ | -2.71 |
| $Ca^+(aq) + 2e^-$ | → | $Ca(s)$ | -2.87 |
| $K^+(aq) + e^-$ | → | $K(s)$ | -2.94 |

*difference in reduction potentials give the emf of an electrochemical cell which utilises the selected half-reactions*

1.56 V

1.10 V

Weak oxidising agents

Strong reducing agents

## Predicting the direction of redox reactions

The standard reduction potentials can be used to predict the likelihood of a particular redox reaction proceeding as written.

The E° values for the two half equations are added.

- An overall + E° means that the reaction may proceed as written.
- An overall – E° means that the reaction will not occur as written.
- In general, substances listed high on the list of reduction potentials (on the left of the reaction arrow) will oxidise those listed lower (on the right of the reaction arrow).

## Metals and dilute acids

e.g. 1. Could zinc react with 1.0 mol $L^{-1}$ HCl acid?
The two possible half-equations are:

$$Zn_{(s)} \rightarrow Zn^{2+}_{(aq)} + 2e^- \qquad E° = +0.76\ V$$

$$2H^+_{(aq)} + 2e^- \rightarrow H_{2(g)} \qquad E° = 0.00\ V$$

$$Zn_{(s)} + 2H^+_{(aq)} \rightarrow Zn^{2+}_{(aq)} + H_{2(g)} \qquad E°_{TOTAL} = +0.76\ V$$

The overall $E°_{TOTAL}$ is positive, hence the reaction may proceed. The $2Cl^-_{(aq)} \rightarrow Cl_{2(g)} + 2e^-$ reaction was not considered since its E° is –1.36 V and the $H^+$ will reduce in preference.

e.g. 2. Could copper react with 1.0 mol $L^{-1}$ HCl acid?
As above:

$$Cu_{(s)} \rightarrow Cu^{2+}_{(aq)} + 2e^- \qquad E° = -0.34\ V$$

$$2H^+_{(aq)} + 2e^- \rightarrow H_{2(g)} \qquad E° = 0.00\ V$$

$$Cu_{(s)} + 2H^+_{(aq)} \rightarrow Cu^{2+}_{(aq)} + H_{2(g)} \qquad E°_{TOTAL} = -0.34\ V$$

Reaction will not proceed since overall E° is negative.

e.g. 3. Magnesium metal is reacted with 1.0 mol $L^{-1}$ nitric acid. Predict the reaction using E° values listed on page 59.

In this example the $Mg_{(s)}$ may be oxidised by $H^+$ or $NO_3^-$.

$$Mg_{(s)} \rightarrow Mg^{2+}_{(aq)} + 2e^- \qquad E° = +2.36\ V$$

$$2H^+_{(aq)} + 2e^- \rightarrow H_{2(g)} \qquad E° = 0.00\ V$$

or

$$NO_3^-{}_{(s)} + 4H^+_{(aq)} + 3e^- \rightarrow NO_{(g)} + 2H_2O_{(l)} \qquad E° = +0.96\ V$$

$$3Mg_{(s)} + 2NO_3^-{}_{(aq)} + 8H^+_{(aq)} \rightarrow 3Mg^{2+}_{(aq)} + 2NO_{(g)} + 4H_2O_{(l)} \qquad E°_{TOTAL} = +3.32\ V$$

Hence $NO_{(g)}$ is produced instead of $H_{2(g)}$ since the reduction potential for $NO_{(g)}$ is higher (i.e. $NO_3^-/H^+$ is a stronger oxidant than $H^+$)

### Question 3.11

Copper will not react with dilute 1.0 mol $L^{-1}$ HCl acid but it will react with dilute 1.0 mol $L^{-1}$ $HNO_3$ acid. Use E° values to determine half equations involved and overall reaction.

## Metals and metal ions – Displacement reactions

Metal displacement reactions will occur if a more reactive metal is placed in a solution of a less reactive metal.

K Na Mg Al Zn Fe Pb $H_2$ Cu Ag Au

reactive ⟶ inert

(metal easily oxidised) (ion easily reduced)

For example, $Mg_{(s)}$ will react with $CuSO_{4(aq)}$ but $Cu_{(s)}$ will not react with $MgSO_{4(aq)}$.

e.g. 4. A strip of zinc metal is added to an aluminium nitrate solution. Predict the reaction, if any.

The two possible half-equations are:

| | | | | | |
|---|---|---|---|---|---|
| $(Zn_{(s)}$ | $\rightarrow$ | $Zn^{2+}{}_{(aq)} + 2e^-) \times 3$ | E° | = | +0.76 V |
| $(Al^{3+}{}_{(aq)} + 3e^-$ | $\rightarrow$ | $Al_{(s)}) \times 2$ | E° | = | -1.66 V |
| $3Zn_{(s)} + 2Al^{3+}{}_{(aq)}$ | $\rightarrow$ | $3Zn^{2+}{}_{(aq)} + 2Al_{(s)}$ | $E°_{TOTAL}$ | = | -0.90 V |

This reaction will not proceed. The reverse reaction, of course, would be favourable. That is, aluminium metal placed in zinc nitrate solution would dissolve. A metal will displace the ions of a less active metal.

## Halogens and halide ions – Displacement reactions

All the halogens are good oxidising agents with fluorine being the strongest. The trend in oxidising ability is related to their position on the periodic table.

$F_{2(g)}$ > $Cl_{2(g)}$ > $Br_{2(l)}$ > $I_{2(s)}$

strong oxidant ⟶ weak oxidant

This means, for example, that $F_{2(g)}$ will displace the halide ions (e.g. $Cl^-$) of any of the other halogens.

e.g. 5. Predict the reaction between $Cl_{2(g)}$ and a solution of $KBr_{(aq)}$.
The two possible equations are:

| | | | | | |
|---|---|---|---|---|---|
| $(Cl_{2(g)} + 2e^-$ | $\rightarrow$ | $2Cl^-{}_{(aq)})$ | E° | = | +1.36 V |
| $2Br^-{}_{(aq)}$ | $\rightarrow$ | $Br_{2(l)} + 2e^-$ | E° | = | -1.08 V |
| $Cl_{2(g)} + 2Br^-{}_{(aq)}$ | $\rightarrow$ | $2Cl^-{}_{(aq)} + Br_{2(l)}$ | $E°_{TOTAL}$ | = | +0.28 V |

Reaction does proceed. The bromide ion will be oxidised by the $Cl_{2(g)}$.

### Question 3.12

A storage plant designer needs to determine if a steel tank can be used to store copper (II) sulfate solution. Use E° values to find out if this is a good idea.

______________________________________________

______________________________________________

______________________________________________

## Limitations of the use of E° tables

The table of standard reduction potentials is useful in predicting reaction tendency and the emf of electrochemical cells.

However they do have limitations:

- The values of E° depend on concentration (1.0 mol $L^{-1}$ solutions)

  e.g. 1.0 mol $L^{-1}$ $HNO_3$ on $Cu_{(s)}$ gives $NO_{(g)}$

  conc. $HNO_3$ on $Cu_{(s)}$ gives $NO_{2(g)}$

- It applies only to aqueous solutions.
- The emf of a cell can depend on temperature, pressure and acidity.
- The E° values give no indication of likely reaction rate.

### Question 3.13

Use standard reduction potentials to determine which of the following reactions are likely to proceed. Assume solutions are 1.0 mol $L^{-1}$.

(a) Iron nails are placed in zinc sulfate solution.

(b) Copper is placed in a nitric acid solution.

(c) Bromine is added to sodium chloride solution.

(d) Chlorine gas is bubbled into hydrogen sulfide solution.

(e) Potassium iodide is added to acidified potassium permanganate solution.

(f) Calcium metal is added to water.

(g) Acidified potassium permanganate solution is added to a solution of iron (II) sulfate.

## Common oxidising and reducing agents

Complete the following tables showing the appropriate half equations and E° values.

### Question 3.14

| Common Oxidising Agents | Name | Reduction Half Equation | E° |
|---|---|---|---|
| $Cl_2$ | | | |
| $MnO_4^-$ (acidified) | | | |
| $Cr_2O_7^{2-}$ (acidified) | | | |
| $ClO^-$ | | | |
| $H^+$ | | | |

Question 3.15

| Common Reducing Agents | Name | Oxidation Half Equation | E° |
|---|---|---|---|
| Mg | | | |
| Zn | | | |
| $C_2O_4^{2-}$ | | | |
| $H_2$ | | | |
| $Fe^{2+}$ | | | |

## 3.5 ELECTROCHEMICAL CELLS

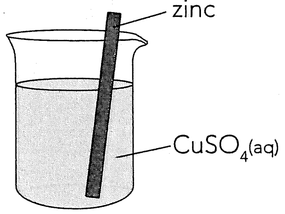
Electrochemical cells are an excellent example of a redox process put to commercial use. The two distinct types of electrochemical cells are *galvanic cells*, such as those used to power a torch or laptop, and *electrolytic cells*, which are typically used for the electrolysis of solutions.

Galvanic cells produce an electric current due to the competition for electrons between two different materials. The reaction is spontaneous. Electrolytic cells require an electrical current to produce a chemical reaction that would not otherwise occur. The process is the reverse of what occurs in a galvanic cell.

In both types of cells a transfer of electrons is involved, that is, a redox reaction. The example in the question below will help to illustrate a typical redox reaction which can be utilized in an electrochemical cell.

Question 3.16

A typical redox reaction is that which occurs when zinc metal is placed in a copper (II) sulfate solution.




(a) Give the two half-equations for this reaction.

oxidation ______________________________

reduction ______________________________

redox equation ______________________________

(b) Describe two changes (observations) you would notice after a few minutes.

(i) ______________________________

(ii) ______________________________

(c) Which species has

(i) gained electrons? ______________________________

(ii) lost electrons? ______________________________

## Galvanic cells

In the reaction discussed earlier (Q 3.16), the zinc metal is in direct contact with the copper (II) ions and hence electrons are transferred at the surface of the zinc metal. In this process, the chemical energy due to the redox reaction is simply given up as heat energy to the surroundings.

In a galvanic cell, such as that illustrated below, the reactants are separated and electrons are forced along an external path. The same spontaneous redox reaction takes place; however, the chemical energy from the reaction produces an external electric current.

Alessandro Volta constructed the first galvanic cell, sometimes also referred to as a voltaic cell or voltaic pile in 1799. This consisted of a series of alternating copper and zinc discs arranged as a column and separated by felt spacers soaked in salt water. Although this first galvanic cell was not so effective over long periods of time it was the first battery to provide a steady continuous flow of electricity.

Improved versions of the galvanic cell were developed during the 19th century such as that devised by John Daniell in 1836. The Daniell cell consisted of a copper can filled with copper sulphate solution in which was immersed a porous pot containing a zinc rod in a zinc sulphate solution. The porous pot prevented the solutions from mixing but allowed ions to flow between the solutions. If the zinc rod and copper can were connected by a wire then an electrical current would flow through the wire.

In the laboratory it is simpler to construct the Daniell cell using two beakers and a salt bridge as shown below. In the cell shown, zinc is oxidized and electrons flow through the external circuit to the copper. Copper ions in the $Cu/Cu^{+2}$ cell are reduced. The salt bridge, which can simply be filter paper soaked in ammonium nitrate solution, allows ions to migrate between the two half cells and so that each cell remains electrically neutral. The choice of salt used in the salt bridge is such as to ensure no precipitation.

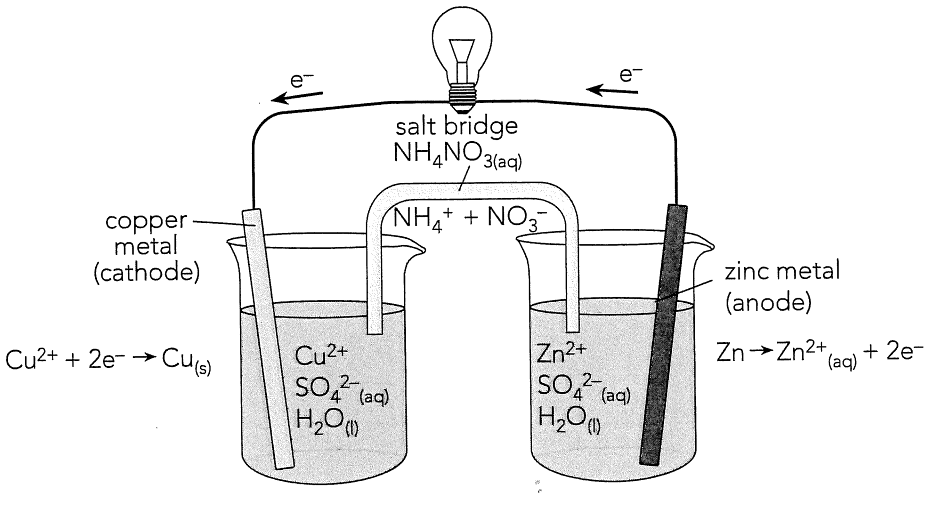



*Figure 3.2 A laboratory version of the Daniell cell consisting of two half-cells, $Zn/Zn^{2+}$ and $Cu/Cu^{2+}$.*

## Question 3.17

Refer to the Daniell cell in Figure 3.2 and answer the following.

(a) Why is a salt bridge necessary?

_______________________________________________

(b) Mark on the diagram the movement of ions in the salt bridge. Show:

- which ions move away from the $Zn/Zn^{2+}$ half-cell.
- which ions move away from the $Cu/Cu^{2+}$ half-cell.

(c) Describe the changes that would occur to:

- the zinc electrode

_______________________________________________

- the copper electrode

_______________________________________________

- the colour of the solution in the $Zn/Zn^{2+}$ cell

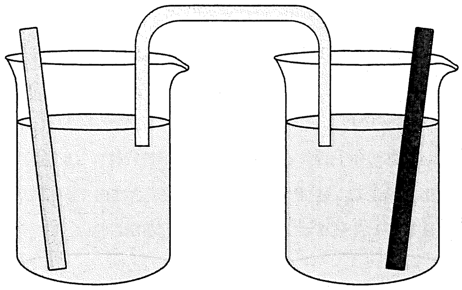
_______________________________________________

- the colour of the solution in the $Cu/Cu^{2+}$ cell

_______________________________________________

## Question 3.18

(a) Complete and fully label the diagram so as to represent a $Mg/Mg^{2+}$, $Cu/Cu^{2+}$ galvanic cell. Label anode, cathode, ions, $e^-$ movement, ion movement, etc.



(b) Anode reaction _______________________

Cathode reaction _______________________

Redox reaction _______________________

(c) What is the expected emf of this cell (1.0 mol $L^{-1}$ solution)?

(d) Eventually this cell will go flat (i.e. stop providing electrons). Detail two conditions that may cause this to happen.

(i) _______________________

(ii) _______________________

## Determining cell voltage

The emf (voltage) produced by a cell is easily determined using standard reduction potentials (i.e. E° values). Note that these apply to 1.0 mol $L^{-1}$ solutions, 101.3 kPa and 25°C.

e.g. for Daniell cell shown in Figure 3.2.

| | | | | | |
|---|---|---|---|---|---|
| $Zn_{(s)}$ | $\rightarrow$ | $Zn^{2+}_{(aq)}$ + 2e- | E° | = | +0.76 V |
| $Cu^{2+}_{(aq)}$ + 2e- | $\rightarrow$ | $Cu_{(s)}$ | E° | = | +0.34 V |
| $Zn_{(s)}$ + $Cu^{2+}_{(aq)}$ | $\rightarrow$ | $Zn^{2+}_{(aq)}$ + $Cu_{(s)}$ | emf | = | 1.10 V |

## Commercial cells – the dry cell*

The dry cell is fairly inexpensive and quite portable. Its compact size allows for a great many uses such as in torches, portable radios and cameras. The dry cell, or **Leclanché cell**, consists of:

- an outer case of zinc (anode )
- a carbon rod surrounded by a paste of $MnO_{2(s)}$ (cathode)
- an electrolyte paste of $NH_4Cl$ and $ZnCl_2$.

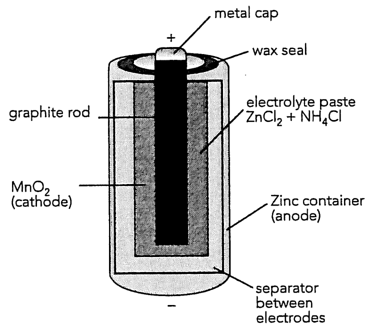



*Figure 3.3 The modern dry cell. Its maximum voltage is 1.48 V.*

Anode reaction $Zn_{(s)} \rightarrow Zn^{2+}_{(aq)} + 2e-$

Cathode reaction $2MnO_{2(s)} + 2H^+_{(g)} + 2e- \rightarrow Mn_2O_{3(s)} + H_2O_{(l)}$

Overall reaction ______________________
*(complete this)*

### Question 3.19

(a) The $H^+_{(aq)}$ ions required in the cathode reaction of the dry cell are provided by the $NH_4^+{}_{(aq)}$ in the electrolyte paste. Write an equation to show how this occurs.

______________________

(b) Why is the Leclanché cell referred to as a primary cell?

______________________

(c) List the major advantages and disadvantages of this type of cell.

Advantages ______________________

______________________

Disadvantages ______________________

______________________

* *The commercial cells discussed in this book provide important examples of redox reactions. However their detailed knowledge may not be required for your course.*

Question 3.20

Alkaline dry cells are also commercially available and are better performers than the acid form. An important example is the silver oxide/zinc cell commonly used in watches and calculators.

The zinc is oxidised to $Zn^{2+}(aq)$ while the silver oxide is reduced to silver. The silver oxide reaction is given. Complete and determine the overall equation.

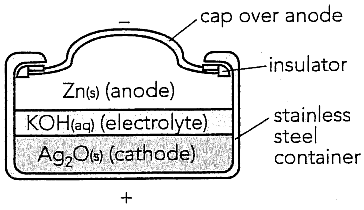



*Figure 3.4 The silver oxide/ zinc cell.*

anode ______________________________

cathode $Ag_2O(s) + H_2O(l) + 2e^- \rightarrow 2Ag(s) + 2OH^-(aq)$

overall ______________________________

## The lead-acid accumulator

Lead-acid batteries are relatively inexpensive and can store large quantities of charge. Although a little bulky, their major advantage is that they can be recharged. They are widely used in motor vehicles and isolated farmhouses. Important features of the lead-acid battery are:

- electrodes
  - anode – spongy lead - large surface area
  - cathode – lead (IV) oxide packed on a metal grid
- electrolyte is concentrated sulfuric acid (4.5 mol $L^{-1}$)
- a typical car battery consists of 6 cells placed in series (12 V total)
- it can be recharged - hence referred to as a secondary cell
- the density of the electrolyte indicates the state of charge.

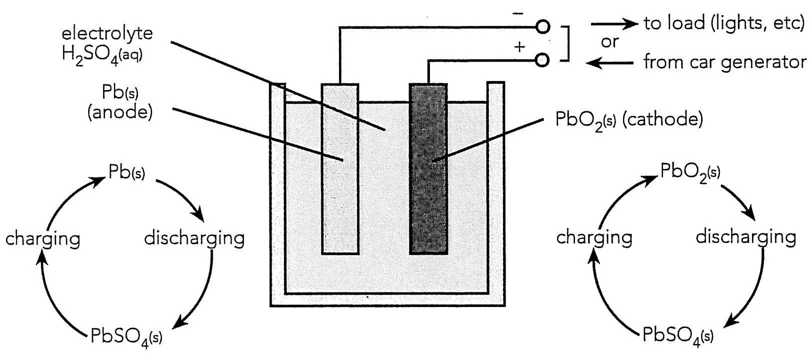



*Figure 3.5 Simplified view of a lead-acid accumulator showing a single cell.*

- Car batteries consist of 6 cells, each providing 2 V.
- The battery is continually being recharged by the car generator (or alternator).

The electrode reactions for *discharge* - that is for when the battery is being used - are as follows:

anode $Pb(s) + SO_4^{2-}(aq) \rightarrow PbSO_4(s) + 2e-$

cathode $PbO_2(s) + 4H^+(aq) + SO_4^{2-}(aq) + 2e- \rightarrow PbSO_4(s) + 2H_2O(l)$

Overall ______________________________
*(complete this)*

## Question 3.21

Write the overall equation for the charging of a lead-acid accumulator.

______________________________________________

## Question 3.22

When an accumulator becomes flat, both electrodes become $PbSO_{4(s)}$.

(i) Explain why no emf would be possible under these conditions.

______________________________________________

(ii) What other change occurs which would more easily help us to decide if the battery is flat?

______________________________________________

______________________________________________

# The fuel cell

A fuel cell is different from both primary and secondary cells in that it does **not** store the reactants or products. Rather, a fuel cell converts the energy of a chemical reaction directly and continuously into electrical energy.

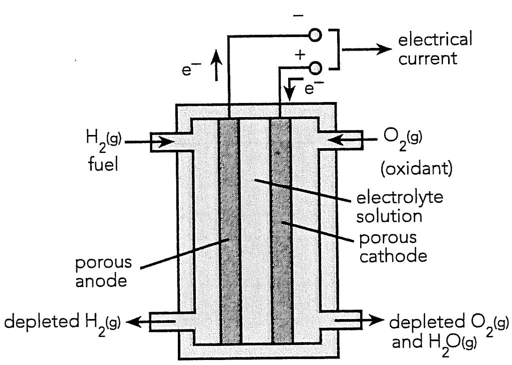



*Figure 3.6 Hydrogen-oxygen fuel cell.*

Important features of a fuel cell are:

- electrodes consist of porous platinum or graphite
- electrolytes can be either basic or acidic
- reactants are usually gaseous (e.g. $H_{2(g)}$, $O_{2(g)}$)
- they supply electricity as reactants are fed into them.

The major advantages of fuel cells are their high efficiency of energy conversion and their ability to generate electricity continuously for a very long time.

Fuel cells are not widely used because of their high cost and slow rate of reaction.

If the electrolyte is acidic the electrode reactions are:

| | |
|---|---|
| anode | $H_{2(g)} \rightarrow 2H^+_{(aq)} + 2e^-$ |
| cathode | $O_{2(g)} + 4H^+_{(aq)} + 4e^- \rightarrow 2H_2O_{(l)}$ |
| overall *(complete this)* | ______________________________ |

## Question 3.23

(a) Use E° values to determine the theoretical emf of the above cell (assume 1.0 mol $L^{-1}$ solution).

_______________________________________________

(b) Write the half-equations for this cell in basic solution.

anode _______________________________________________

cathode _______________________________________________

(c) What is the overall reaction in both cases?

_______________________________________________

(d) How does this overall reaction compare with the combustion of hydrogen gas?

_______________________________________________

## Question 3.24

A fuel cell is designed to use methane ($CH_4$) and oxygen in a similar manner to the $H_2/O_2$ fuel cell. If the electrolyte is acidic, determine the electrode reactions.

anode _______________________________________________

cathode _______________________________________________

overall _______________________________________________

## 3.6 ELECTROLYSIS

Electrolysis refers to the chemical changes that take place when an electrical current is passed through a molten substance or solution. Electrolysis occurs in an electrolytic cell and is the opposite process to that which occurs in a galvanic cell.

The reactions that occur during electrolysis are *forced reactions*, that is, they are not spontaneous reactions. By applying a suitable voltage, electrical energy is converted to chemical energy. A comparison of the processes that occur in galvanic cells and electrolytic cells is shown below.

| GALVANIC CELL | ELECTROLYTIC CELL |
|---|---|
| chemical energy in | electrical energy in |
| ↓ | ↓ |
| spontaneous chemical reaction | forced chemical reaction |
| ↓ | ↓ |
| electrical energy out | chemical energy out |

*Figure 3.7 Comparing galvanic and electrolytic cells. Galvanic cells produce electrical energy due to spontaneous reactions. Electrolytic cells have an applied external voltage which causes forced reactions to occur (electrolysis). In both cases the reactions are redox reactions.*

Electrolysis has many important industrial applications. Common examples are listed below. Some of these applications will be discussed in more detail later in this chapter.

| ELECTROLYSIS APPLICATION | COMMON EXAMPLE |
| --- | --- |
| The extraction of metals | Aluminium metal from alumina ($Al_2O_3$) |
| Electrorefining | Pure copper from blister copper |
| Electroplating | Chrome plating, silver plating |
| Anodising | Corrosion resistant $Al_2O_3$ layer on Al metal |
| Chemical manufacture | Production of Na, NaOH, $Cl_2$, $H_2$ |
| Recharging batteries | Car batteries, nickel-cadmium batteries |

## Electrolytic cells

Systems that bring about electrolysis reactions are called electrolytic cells. A typical example is shown below. An applied external voltage causes reactions at the two electrodes.

ANODE: Where oxidation occurs
CATHODE: Where reduction occurs
*This is the same for all electrochemical cells*

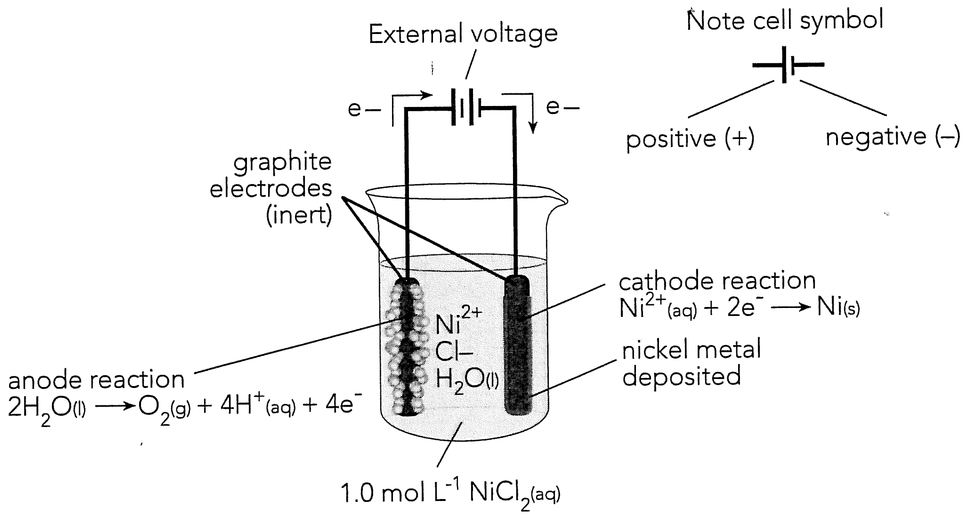



*Figure 3.8 A typical electrolytic cell. An external applied voltage supplies electrons to the cathode where a reduction reaction takes place. At the anode, an oxidation reaction occurs which results in the same number of electrons being produced as those consumed at the cathode. Note that only ions flow through the solution. As they do, they complete the electrical circuit.*

## Predicting electrode reactions

Often, more than one reaction is possible at each electrode of an electrolytic cell. For each electrode we need to consider all substances present and whether the electrode is inert or reactive. We can to use E° values to predict reactions and voltages required. The concentration of solutions can also have a marked effect on E values and products formed. E° values apply only to 1.0 M solutions. For simplicity here we assume all 1.0 M solutions and inert electrodes. We discuss the effects of electrodes and concentration later below.

Consider the electrolytic cell shown above in Figure 3.8

*Possible cathode reactions. Note that only reduction reactions can occur here:*

$Ni^{2+}_{(aq)} + 2e^- \rightarrow Ni_{(s)}$ E° = –0.24 V ✓*This reaction is favoured*
$2H_2O_{(l)} + 2e^- \rightarrow N_{2(g)} + 2OH^-_{(aq)}$ E° = –0.83 V

*Possible anode reactions. Note that only oxidation reactions can occur here:*

| | |
|---|---|
| $2Cl^-_{(aq)} \rightarrow Cl_{2(g)} + 2e^-$ | $E° = -1.36$ V |
| $2H_2O_{(l)} \rightarrow O_{2(g)} + 4H^+_{(aq)} + 4e^-$ | $E° = -1.23$ V ✓*This reaction is favoured* |

*Hence overall reaction:*

$$2Ni^{2+}_{(aq)} + 2H_2O_{(l)} \rightarrow 2Ni_{(s)} + O_{2(g)} + 4H^+_{(aq)} \qquad E° = -1.49 \text{ V}$$

*Note*

- The least negative half reaction is always favoured.
- A minimum voltage of + 1.49 V must be applied to cause a reaction.
- Graphite electrodes can be considered inert and do not react.

## Using reactive electrodes

Where reactive electrodes are used, they need to be considered in predicting reactions. If nickel metal electrodes were used in the previous example with 1.0 mol $L^{-1}$ $NiCl_{2(aq)}$ we would have:

*Possible cathode reactions:*

| | |
|---|---|
| $Ni^{2+}_{(aq)} + 2e^- \rightarrow Ni_{(s)}$ | $E° = -0.26$ V ✓*This reaction is favoured* |
| $2H_2O_{(l)} + 2e^- \rightarrow H_{2(g)} + 2OH^-_{(aq)}$ | $E° = -0.83$ V |

*Possible anode reactions:*

| | |
|---|---|
| $Ni_{(s)} \rightarrow Ni^{2+}_{(aq)} + 2e^-$ | $E° = +0.26$ V ✓*This reaction is favoured* |
| $2Cl^-_{(aq)} \rightarrow Cl_{2(g)} + 2e^-$ | $E° = -1.36$ V |
| $2H_2O_{(l)} \rightarrow O_{2(g)} + 4H^+_{(aq)} + 4e^-$ | $E° = -1.23$ V |

*Hence overall reaction:*

| | | |
|---|---|---|
| $Ni_{(s)}$ | $\rightarrow$ | $Ni_{(s)}$ |
| (at the anode) | | (at the cathode) |

We can see that by using nickel electrodes one of the preferred reactions is different from the previous example. At the anode nickel metal reacts rather than water. Effectively, nickel metal is oxidized at the anode and the resulting ions are reduced back to the metal at the cathode. This type of reaction is very important in electroplating and the electrorefining of metals like copper.

### Question 3.25

(a) If the nickel electrodes in the previous example were replaced with copper electrodes, the overall reaction would be different. Consider the possible electrode reactions to determine the overall result.

Possible cathode reactions:

________________________________________

________________________________________

Possible anode reactions:

________________________________________

________________________________________

Overall reaction:

___

___

(b) The reactions within this electrolytic cell may change after some time. Explain why.

___

___

___

___

## Aqueous solutions – concentration effect

The concentration of an aqueous solution can affect the preferred reactions that occur in an electrolytic cell. E° values apply only to 1.0 M solutions. At other concentrations the E values change and other competing reactions may be more favoured.

In all aqueous solutions water is always a possible reactant; at either electrode. In fact if a low concentration solution of say NaCl or $H_2SO_4$ is used, then *electrolysis of water* will occur. The products would be oxygen gas at the anode and hydrogen gas at the cathode.

Let us consider the electrolysis of a NaCl solution at different concentrations. Assume inert electrodes. The possible electrode reactions and E° values (1.0 M solutions) are listed. For neutral solutions the (non standard) E value for the water reactions is also given.

*Possible cathode reactions:*

$Na^+_{(aq)} + e^- \rightarrow Na_{(s)}$ E° = -2.71 V

$2H_2O_{(l)} + 2e^- \rightarrow H_{2(g)} + 2OH^-_{(aq)}$ E° = –0.83 V *(E = –0.41 V at pH = 7)*

*Possible anode reactions:*

$2Cl^-_{(aq)} \rightarrow Cl_{2(g)} + 2e^-$ E° = -1.36 V

$2H_2O_{(l)} \rightarrow O_{2(g)} + 4H^+_{(aq)} + 4e^-$ E° = -1.23 V (*E = -0.82 V at pH = 7*)

*Aqueous NaCl (1.0 M solution)*. At the cathode, water is reduced rather than the sodium ions as indicated by the E° values. Hydrogen gas is the only product at both high and low concentrations of solution. At the anode we can see that for 1.0 M solutions the E° values for the possible reactions are very similar. Oxygen gas is produced in preference to chlorine gas although some chlorine gas may also be produced. At very low concentration only oxygen gas is produced.

*Aqueous NaCl (concentrated solution)*. At high concentrations only chlorine gas is produced at the anode. Hydrogen gas is still produced at the cathode. The electrolysis of a concentrated aqueous solution of sodium chloride is an important industrial process.

The *Nelson cell* is used to electrolyse a concentrated brine solution to produce hydrogen, chlorine and sodium hydroxide, all important industrial chemicals. The hydrogen and chlorine are produced as discussed above and the sodium hydroxide is the result of the hydroxide ions produced at the cathode combining with the sodium ions in solution. Essentially the $NaCl_{(aq)}$ solution becomes a $NaOH_{(aq)}$ solution.

## Question 3.26

Consider the electrolysis of sodium chloride solutions at different concentrations as discussed above. Assuming inert electrodes give the electrode reactions and list the products formed for the following concentrations.

(a) Aqueous NaCl (1.0 mol $L^{-1}$)

Anode reaction ________________________________

Cathode reaction ________________________________

Products are ________________________________

(b) Aqueous NaCl (concentrated solution)

Anode reaction ________________________________

Cathode reaction ________________________________

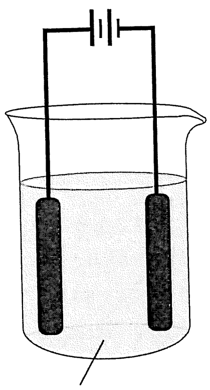
Products are ________________________________

## Question 3.27

A 1.0 mol $L^{-1}$ solution of $HCl_{(aq)}$ is to be electrolysed using inert electrodes as shown.

(a) Fully label the diagram showing:

- species in the beaker
- electron flow
- anode and cathode




(b) Determine possible reactions at each electrode and give the overall reaction. Include E° values.

anode ________________________________

________________________________

cathode ________________________________

________________________________

overall ________________________________

________________________________

## Electrolysis of molten salts

As we have seen above, the electrolysis of aqueous solutions often involves the reaction of water itself. Solid ionic salts do not conduct electricity. However if they are heated sufficiently they become molten and can be electrolysed. In this situation the ions of the ionic salt will react at the electrodes. For example, if sodium chloride is heated to above 800°C it melts and can be electrolysed as shown below.

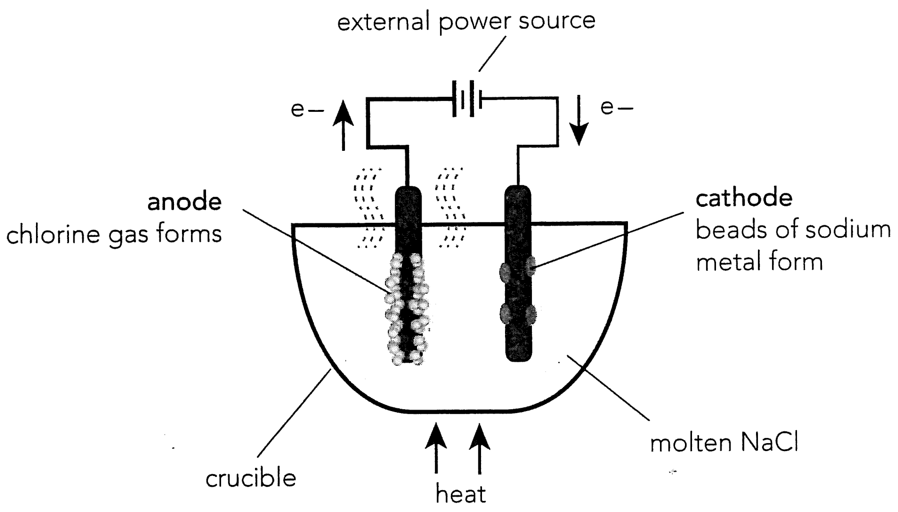



*Figure 3.9 Electrolysis of molten sodium chloride. Only sodium and chloride ions are available to react at the inert electrodes. Sodium liquid is produced at the cathode and chlorine gas at the anode.*

### Question 3.28

Consider the electrolysis of molten sodium chloride as discussed above. Assuming inert electrodes, give the electrode reactions for the following examples.

(a) Molten NaCl

Anode reaction ________________________________

Cathode reaction ________________________________

Products are ________________________________

(b) Molten $PbI_2$

Anode reaction ________________________________

Cathode reaction ________________________________

Products are ________________________________

## Electroplating

Metals may be electroplated by adding a very thin layer of another metal by the process of electrolysis. This important industrial process is used in order to improve the appearance of more base metals such as iron and also improve their resistance to corrosion.

Silver plating, for example, can enhance the looks of items such as bracelets, necklaces, cutlery and ornamental objects. A very thin layer of silver is coated on the object by attaching them to the cathode of an electrolytic cell as shown. The finished product will have the look of silver but cost much less than if the item was solid silver.

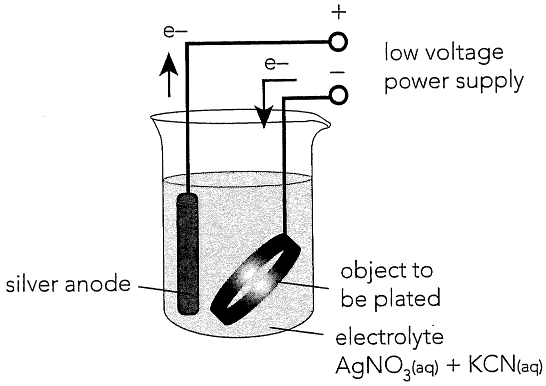



*Figure 3.10 Electroplating with silver. The metal object to be plated is always attached to the cathode. The anode is made of the metal used for the coating. It does not need to be pure.*

Gold similarly can be used for electroplating items of jewellery. It is also used in the electronics industry in a similar way to produce tarnish resistant contacts and switching gear. Other important examples of electroplating are tin plating, zinc plating and chrome plated steel.

### Question 3.29

For the electroplating of silver as shown above, give the relevant reactions.

anode ____________________________________

cathode ____________________________________

overall ____________________________________

### Question 3.30

To get a really smooth finish when electroplating with silver, the silver ion concentration in solution must be very low. This is why KCN is added to the electrolyte solution. An equilibrium is established as follows:

$$Ag^{+}_{(aq)} + 2CN^{-}_{(aq)} \rightleftharpoons Ag(CN)_2{}^{-}_{(aq)}$$

Explain how this equilibrium reaction helps.

____________________________________________

____________________________________________

## Electrolytic refining of copper

Copper is produced from its ores by a series of processes. The final smelt contains about 98.5% of copper. However the purity required for electrical wiring, one of the main uses of copper, is very high (≈ 99.9%). To achieve this high purity, crude copper or 'blister' copper is refined as shown below.

This electrolytic process is very similar to electroplating. The copper and other reactive metals within the anode are oxidised and become part of the electrolyte solution. The high cost of electrorefining is partly offset by the recovery of valuable impurities such as silver and gold. The use of a very low voltage, approximately 0.3 V, ensures that only copper is deposited at the cathode and that the less active silver and gold are not oxidised at the cathode but are left behind as part of the anode mud.

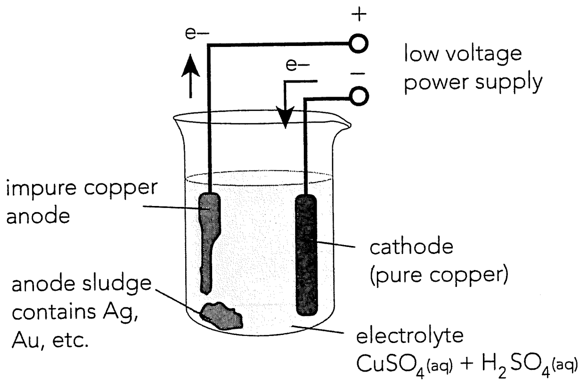



*Figure 3.11 Electrolytic refining of copper. Industrially, copper anodes consist of large copper slabs produced from crude copper (≈ 98.5%) or recycled high grade copper scrap. Impurities mostly include gold, silver, lead and nickel. These impurities are recovered from the anode mud or from solution.*

Question 3.31

(a) For the electrorefining of copper as shown above, give the relevant reactions.

anode ________________________________

cathode ________________________________

overall ________________________________

(b) The gold and silver impurities not oxidised at the anode simply fall to the bottom and become part of the anode *sludge* or *slime*.

(i) Explain why this is the case. Refer to E° values in your answer.

(ii) Of what use might the anode mud be?

(c) Oxidised metal impurities such as $Ni^{2+}$ and $Pb^{2+}$ are not deposited at the cathode.

(i) Explain why this is the case. Refer to E° values in each case.

(ii) Suggest how the nickel and copper ions may be removed or recovered.

## 3.7 CORROSION OF METALS

Corrosion is an electrochemical process that occurs when metals are oxidised by substances in their environment.

Anode (metal oxidises): Metal → (Metal ions) $^{x+}$ + x e-

Cathode: Four common, but not exhaustive, reduction processes are given below:

| | | |
|---|---|---|
| (i) | $2H^+ + 2e^- \rightarrow H_2$ | metal exposed to moist, acidic conditions |
| (ii) | $O_2 + 2H_2O + 4e^- \rightarrow 4OH^-$ | metal exposed to moisture and oxygen (very common) |
| (iii) | $2H_2O + 2e^- \rightarrow H_2 + 2OH^-$ | metal exposed to moisture but a low concentration of oxygen, e.g. shovel blade in the soil |
| (iv) | $(\text{Metal ions})^{x+} + x\ e^- \rightarrow \text{Metal}$ | metal in contact with another metal with a greater reduction potential |

Preventing corrosion is often about limiting the contact of the metal with these four processes.

### Question 3.32

List the following metals in order of increasing tendency to be oxidised.

Fe, Zn, Au, Mg, Na, Ag, Al, Pb, Cu

____________________________________________

When some metals oxidise, they form a tough, protective oxide coating that greatly limits the ability of the oxidising agent to come in contact with the metal. Aluminium is a common example. Even though Al will oxidise rapidly, the oxide coating protects the underlying Al from further corrosion. Iron, however forms a oxide coating that is easily penetrated by $O_2$ and $H_2O$ and subsequently the oxide layer does not protect the Fe from further corrosion.

Rust – A Special Example of Corrosion

The most common and important example of corrosion is the oxidation of iron. This is caused by the action of oxygen and water vapour in the air.

The formation of rust involves a series of reactions.

- *The initial oxidation of the iron.*

$$Fe_{(s)} \rightarrow Fe^{2+}_{(aq)} + 2e^- \quad \text{(anodic reaction)}$$
$$O_{2(g)} + 2H_2O_{(l)} + 4e^- \rightarrow 4OH^-_{(aq)} \quad \text{(cathodic reaction)}$$

- *The further oxidation of the $Fe(OH)_{2(s)}$ formed.*

$$4Fe(OH)_{2(s)} + 2H_2O_{(l)} + O_{2(g)} \rightarrow 4Fe(OH)_{3(s)}$$

pale green — red brown

- *The partial dehydration of the $Fe(OH)_{3(s)}$ to rust. One possible reaction is shown.*

$$2Fe(OH)_3 \rightarrow Fe_2O_3.H_2O_{(s)} + 2H_2O_{(l)}$$

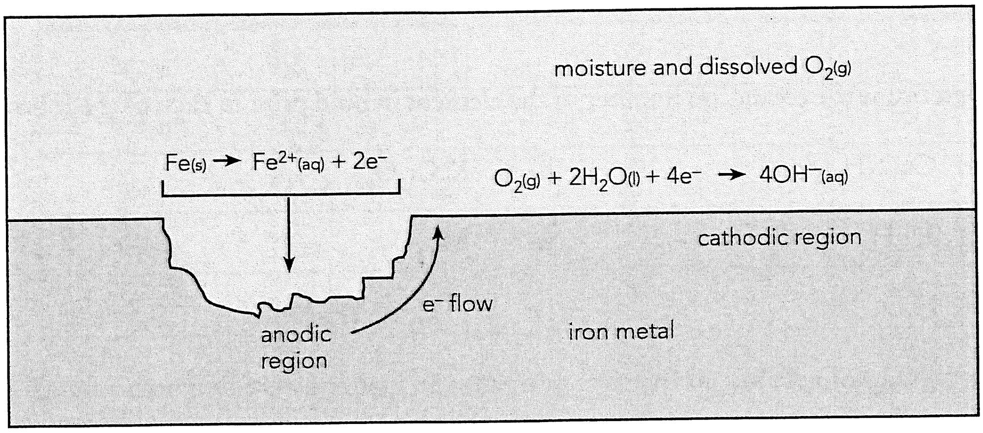



*Figure 3.12 Corrosion of iron by the action of oxygen and water.*

This is an electrochemical process. Areas of stress in the iron become anodic while areas of high oxygen concentration become cathodic. Electrons flow through the metal from the anode to the cathode. The moisture acts as the electrolyte.

## Corrosion prevention

Prevention of corrosion is of tremendous economic importance (future millionaires – here is an area where you can make money). To prevent corrosion we must inhibit the reactions discussed on the previous page. This can be done by:

- excluding air and/or water from the metal surface by
  - protecting the iron surface with paint, grease, plastic
  - plating the iron surface with metals such as chromium or metallic tin
- using a sacrificial anode (more reactive metal) by
  - galvanising iron with zinc – zinc corrodes in preference
  - attaching magnesium and aluminium to ships
- using cathodic protection so that jetties and pipelines are rendered negative by a low voltage dc current. The anodes may be made of scrap iron (e.g. old engine blocks) or titanium coated inert electrodes.

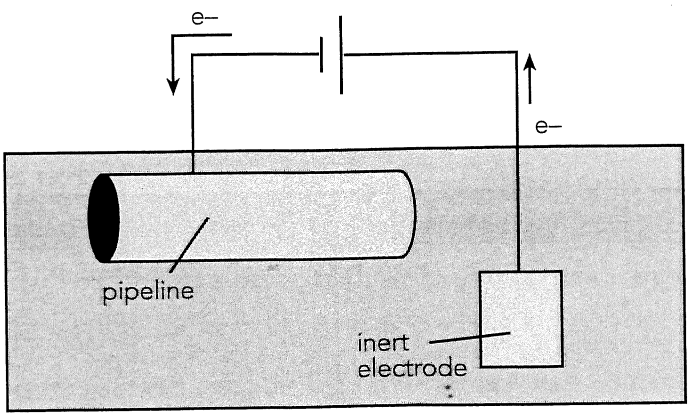



*Figure 3.13 Cathodic protection. The pipeline is made negative to prevent its oxidation. Instead, $H_2O$ is reduced at the pipeline, and oxidised at the inert electrode.*

# REVIEW QUESTIONS

## Chapter 3: Oxidation and Reduction

### Redox

1. Determine the oxidation number of the element in bold print in each of the following:

(a) $\mathbf{Cu}(NH_3)_4^{2+}$ (b) $K_2\mathbf{Cr}_2O_7$

(c) $Mg\mathbf{H}_2$ (d) $\mathbf{Mn}_2O_3$

(e) $H_2\mathbf{O}_2$ (f) $H_2\mathbf{C}_2O_4.2H_2O$

(g) $(NH_4)_2SO_4.\mathbf{Fe}SO_4.6H_2O$

2. Use oxidation numbers to identify the element that has been oxidised (if any has) in each of the following:

(a) $2H_2(g) + O_2(g) \rightarrow 2H_2O(l)$

(b) $2Na(s) + H_2S(g) \rightarrow Na_2S(s) + H_2(g)$

(c) $MnO_2(s) + 4H^+(aq) + 2Cl^-(aq) \rightarrow Mn^{2+}(aq) + Cl_2(g) + 2H_2O(l)$

(d) $AgBr(s) + 2S_2O_3^{2-}(aq) \rightarrow [Ag(S_2O_3)_2]^{3-}(aq) + Br^-(aq)$

(e) $5H_2C_2O_4(aq) + 2MnO_4^-(aq) + 6H^+(aq) \rightarrow 2Mn^{2+}(aq) + 8H_2O(l) + 10CO_2(g)$

3. Use oxidation numbers to identify the element that has been reduced (if any has) in each of the following:

(a) $4Fe(s) + 3O_2(g) \rightarrow 2Fe_2O_3(s)$

(b) $Cu(s) + 4H^+(aq) + 2NO_3^-(aq) \rightarrow Cu^{2+}(aq) + 2H_2O(l) + 2NO_2(aq)$

(c) $4Au(s) + 8CN^-(aq) + 2H_2O(l) + O_2(g) \rightarrow 4Au(CN)_2^-(aq) + 4OH^-(aq)$

(d) $Pb^{2+}(aq) + 2I^-(aq) \rightarrow PbI_2(s)$

4. Complete the following table by classifying the substances listed as **oxidising agents** or as **reducing agents.**

*Oxalic acid, zinc, oxygen gas, hydrogen gas, chlorine gas, carbon, potassium permanganate (acidified), iron (II) ions, concentrated sulfuric acid, potassium dichromate (acidified), hydronium ion, and concentrated nitric acid.*

| Oxidising Agents (oxidants) | Reducing Agents (reductants) |
|---|---|
| | |

5. Complete the following table showing the observations you would expect to make when the following substances are mixed.

| Equation | Colour of Reactants | Colour of Products |
|---|---|---|
| $Br_2(aq)$ + $I^-(aq)$ → | | |
| $Br^-(aq)$ + $Cl_2(aq)$ → | | |
| $I^-(aq)$ + $Cl_2(aq)$ → | | |
| $Ag^+(aq)$ + $Cu(s)$ → | | |
| $Mg(s)$ + $Cu^{2+}(aq)$ → | | |

6. Use oxidation numbers and half equations to balance each of the following equations (assuming acidic conditions).

(a) $S_2O_3^{2-}(s) + I_2(s) \rightarrow S_4O_6^{2-}(aq) + I^-(aq)$

(b) $Br^-(aq) + SO_4^{2-}(aq) \rightarrow SO_2(g) + Br_2(l)$

(c) $Ag_2S(s) + H^+(aq) + NO_3^-(aq) \rightarrow Ag^+(aq) + NO(g) + S(s) + H_2O(l)$

(d) $CrO_4^{2-}(aq) + H^+(aq) + Cl^-(aq) \rightarrow Cr^{3+}(aq) + Cl_2(g) + H_2O(l)$

(e) $MnCl_2(s) + OH^-(aq) + Br_2(aq) \rightarrow MnO_2(s) + Cl^-(aq) + Br^-(aq) + H_2O(l)$

7. Use oxidation numbers and half equations to write balanced nett equations for each of the following reactions. You will also need to use your knowledge of solubility rules, etc, to produce the correct **ionic** equations.

(a) Silver metal being dissolved in concentrated nitric acid to produce silver nitrate, nitrogen dioxide and water.
(b) Sulfur powder is thoroughly mixed with a solution of acidified sodium dichromate to produce sodium sulfate and the insoluble chromium (III) oxide.
(c) Hydrogen sulfide gas is bubbled through dilute nitric acid producing sulfur, nitrogen monoxide and water.
(d) Chlorine water is made dissolving bleaching powder, $CaCl_2.Ca(ClO)_2$ in water and then adding some dilute hydrochloric acid. When bleaching powder dissolves in water it forms $Ca(ClO)_2$ solution.
(e) An iron (II) sulfate solution is mixed with an acidified potassium permanganate solution.
(f) A potassium iodide solution is mixed with an acidified solution of potassium dichromate.
(g) A solution of acidified potassium permanganate is mixed with a solution of hydrogen peroxide.

8. Disproportionation is said to occur when an element is simultaneously oxidized and reduced. Show how this can occur with hydrogen peroxide by using appropriate half equations.

9. Sodium hypochlorite and calcium hypochlorite are two sources of the $OCl^-$ ion which is a commonly used oxidising agent. Give two uses of the hypochlorite ion.

10. Rewrite the following list of elements so that they are in increasing order of strength as reducing agents.

*Silver, chlorine, hydrogen, lead, gold and calcium.*

11. $MnO_2$ can be reduced according to the following equation:

$$MnO_{2(s)} + 4H^+{}_{(aq)} + 2e^- \rightarrow Mn^{2+}{}_{(aq)} + 2H_2O_{(l)}$$

Which halide ions could be oxidised by manganese (IV) oxide?

12. Rewrite the following list in order of increasing strength as oxidising agents.

$$H_2O_2\ ,\ Cr^{3+}\ ,\ Br_2\ ,\ Cr_2O_7^{2-}\ ,\ MnO_4^-\ ,\ K^+\ ,\ Au^{3+}.$$

13. (a) Name two substances that will oxidise $Sn^{2+}$ to $Sn^{4+}$.

(b) Name two substances that will reduce $Fe^{3+}$ to $Fe^{2+}$.

(c) Name a substance that will oxidise solid gold.

(d) Name a substance that will oxidise silver but that will not oxidise platinum.

14. What might happen if aluminium rivets are used to fasten zinc coated gutters to a roof?

15. The standard reduction potential for the half cell containing 1.00 mol $L^{-1}$ $Al^{3+}$ ions and Al solid at 25°C and 101.3 kPa pressure is −1.68 V.

(a) Use the $Al^{3+}$/Al half cell (and others if necessary) to explain what is meant by the term **standard reduction potential.**

(b) Is it possible to have this half cell operating in isolation from any other chemical species that may react?

(c) Name two chemical species that could be used to cause $Al^{3+}$ to be reduced to Al solid.

16. Use a table of standard reduction potentials to determine if a reaction is possible between the following chemical species. If a reaction is possible, use half equations to write a balanced net equation.

(a) A strip of Cu is added to a 1 mol $L^{-1}$ HCl solution.

(b) Solutions of copper (I) nitrate and iodine water are mixed.

(c) An acidified potassium permanganate solution is added to a sample of hydrogen peroxide.

(d) A strip of copper metal is placed in a solution of concentrated nitric acid. Use E° values listed on page 59.

(e) A piece of sodium metal was placed into a beaker containing a 1 mol $L^{-1}$ solution of calcium nitrate.

(f) An acidified 1 mol $L^{-1}$ $KMnO_4$ solution is mixed with a 1 mol $L^{-1}$ solution of HCl.

(g) An acidified 1 mol $L^{-1}$ $K_2Cr_2O_7$ solution was mixed with a 1 mol $L^{-1}$ solution of HCl.

(h) A piece of iron was submersed in a recently boiled beaker of pure water.

17. A table of standard reduction potentials would predict that aluminium metal will oxidise when placed into a beaker of water.

i.e. $Al_{(s)} \rightarrow Al^{3+}_{(aq)} + 3e^-$ $\quad E° = +1.68\ V$

$2H_2O_{(l)} + 2e^- \rightarrow H_{2(g)} + 2OH^-_{(aq)}$ $\quad E° = -0.83\ V$

Explain why aluminium window frames do not oxidise on exposure to water.

## Galvanic cells

18. In any galvanic cell:

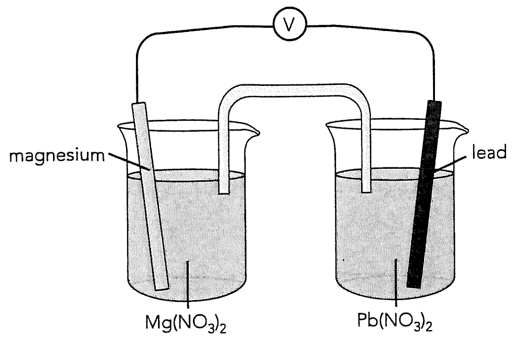
(a) What process occurs at the anode?

(b) What process occurs at the cathode?

(c) What charge is assigned to the anode?

(d) What charge is assigned to the cathode?

19. Consider the following simple galvanic cell:




(a) What direction would electrons flow in the external circuit?

(b) What would be the charge carriers in the external circuit?

(c) What would be the charge carriers in the solutions?

(d) What would be a suitable solution to use in the salt bridge?

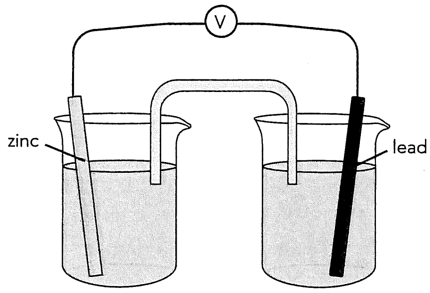
(e) Write the equation to show the oxidation process.

(f) Write the equation to show the reduction process.

(g) If the cell was set up according to standard conditions, what would be the reading on the voltmeter?

(h) Why is a salt bridge necessary in cells that have this style of construction?

20. A galvanic cell was constructed using the following set up:




The solution in both beakers was dilute hydrochloric acid and potassium nitrate was used in the salt bridge.

(a) Would the cell still work if the zinc strip was removed from its beaker and placed into the same beaker as the lead strip (but not touching the lead)?

(b) Would a reaction occur if lead and zinc strips were allowed to touch? How would the reading on the voltmeter be affected?

21. For each of the following combinations, write the equation for the process occurring at the anode and the cathode and predict the cell emf (assume standard conditions).

(a) $\{Zn/Zn^{2+}\}$ and $\{Cu/Cu^{2+}\}$

(b) $\{Ag/Ag^{+}\}$ and $\{Fe/Fe^{2+}\}$

(c) $\{I_2/I^{-}\}$ and $\{Al/Al^{3+}\}$

(d) $\{Br_2/Br^{-}\}$ and $\{F_2/F^{-}\}$

(e) $\{MnO_4^{-}/H^{+}/Mn^{2+}/H_2O\}$ and $\{H_2O_2/O_2/H^{+}\}$

## Electrolysis

22. Complete the following table which compares and contrasts galvanic and electrolytic cells.

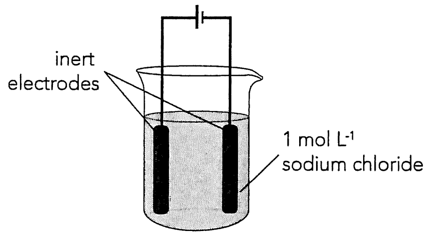
| Galvanic Cells | Electrolytic Cells |
|---|---|
| | Forced chemical reaction |
| Oxidation occurs at the: | |
| | Polarity of anode is: |
| Electrical energy is created by a chemical change. | |

23. Write the oxidation and reduction half equations for the reactions that occur when an electrical current is passed through the following molten salts. Inert electrodes have been used. Give the minimum voltage required to produce the reaction in each case.

(a) sodium chloride (b) zinc bromide

(c) lead (II) iodide (d) potassium fluoride

24. Consider the following electrolytic cell:




(a) Identify all chemical species in this cell.

(b) List those that can be oxidised.

(c) Use the table of standard reduction potentials to identify the species that is most likely to be oxidised.

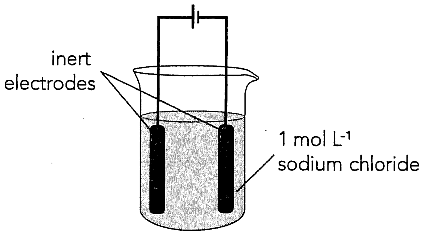
(d) Repeat parts (b) and (c) to identify what species is most likely to be reduced.

(e) Write the half equations and nett equation for the reaction that occurs.

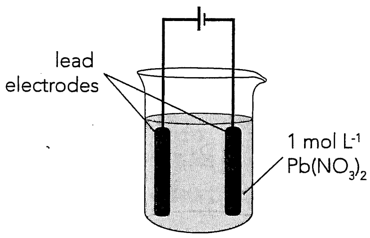
(f) What minimum voltage does the battery need to supply to make the reaction occur?

25. In which of the following cells is water oxidised or reduced?

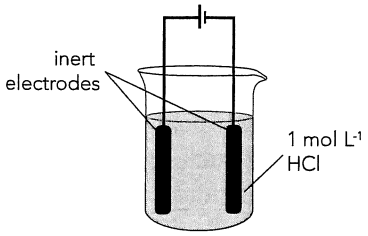
(a)



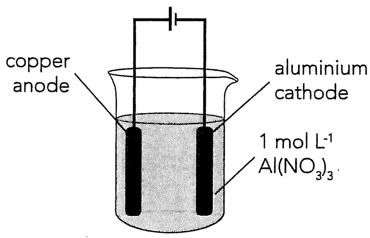

(b)




(c)




(d)




26. Write the oxidation half equation, reduction half equation, nett equation and voltage required for each of the reactions that occur in question 25, parts (a) to (d).

27. With the aid of a labelled diagram and appropriate equations, explain how a steel bracelet could be given a coating of silver. Clearly indicate in your diagram what the cathode; anode and electrolyte are composed of.

28. Two commonly used classroom reagents are 1 mol L$^{-1}$ $H_2SO_4$ and 1 mol L$^{-1}$ NaOH. Write the nett equations for the electrolysis of each of these reagents (individually) if inert electrodes are used.

29. The corrosion of iron to form rust is a complicated process and it is often best to break it down into several simpler stages. Complete the following by filling in the blank spaces in the explanation or in the paragraph.

- Iron corrodes (oxidises) at the anode: $Fe_{(s)} \rightarrow Fe^{2+} + 2e^-$
- For the iron to be oxidised, some other chemical must be ______________.
- If oxygen and water are present at the ______________ then hydroxide ions will form:

$$O_2(g) + 2H_2O(l) + \_\_\_\_\_\_\_\_\_\_\_\_\_\_ \rightarrow 4OH^-(aq)$$

- The $Fe^{2+}$ and $OH^-$ ions form the insoluble ______________ hydroxide.
- The presence of dissolved oxygen in the water will cause the further oxidation of the iron to form a compound called 'rust'.
- Two possible formulae for rust are ______________ and ______________.

30. Using the information from question 29, explain why the addition of an organic base such as ethylene glycol helps to inhibit the corrosion of car engines.

31. Tin is often used to coat iron cans which are subsequently called tin cans.

(a) Why is tin used as a coating for iron?

(b) Use half equations and E° values to explain why this coating will only work while it is coherent and no iron is exposed.

(c) Use half equations and E° values to explain why a zinc coating is much more effective in preventing the corrosion of iron.

32. Solar hot water systems sometimes have a metal rod (magnesium or zinc) inserted into them to act as a sacrificial anode. Explain how this anode would help prevent a steel tank from corroding.

33. To reduce the likelihood of car doors corroding it is often recommended to apply a coating of light oil to the internal parts of the door and to clean the drain holes regularly. Explain how these two procedures help prevent corrosion.

34. Long pipelines are expensive to construct and so minimising maintenance problems after construction is an economic necessity. One such method is to connect the pipeline to a DC electricity supply. The pipeline is made the cathode while pieces of metal at various distances along the pipeline are connected as the anode.

(a) Explain why this procedure would help to reduce the overall cost of pipe maintenance.

(b) What is likely to happen to the metal that is connected as the anode?

35. In the extraction of iron from iron ore in a blast furnace, carbon monoxide is the main reducing agent. The blast furnace is filled with iron ore (e.g. $Fe_2O_3$), coke (carbon) and limestone ($CaCO_3$, $SiO_2$) and heated to temperatures of nearly 2000°C.

(a) How is the carbon monoxide produced in the blast furnace? Give relevant equations.

(b) Write the equation for the reduction of $Fe_2O_3$ to Fe by CO.

# FOR THE EXPERTS

36. A student sets out to build a galvanic cell using the following components:

- Freshly cleaned strips of Al and Sn.
- Solutions (1.0 m) of aluminium nitrate, tin (II) nitrate and potassium nitrate.
- Glassware to construct the cell.
- Electrical leads and a voltmeter.

(a) Draw a labelled diagram to show how these materials could be used to make a galvanic cell. Your diagram must include the materials to be used as the salt bridge, the anode and the cathode.

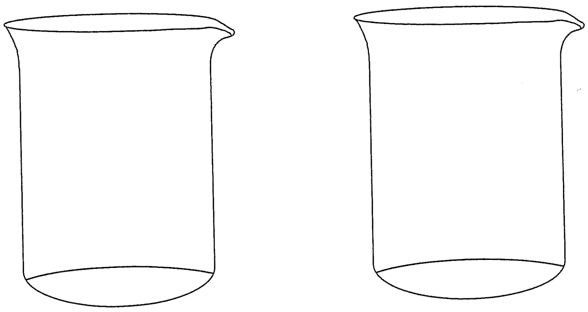
(b) Write equations for the processes occurring at the anode and the cathode.

(c) Predict the emf of the cell (if at standard conditions).

(d) Using chemical symbols and arrows, show the direction of movement of the chemical species in this galvanic cell.

(e) After some time it was noticed that the emf of the cell was lower. Suggest possible reasons for this.

(f) Which of the electrodes is likely to have increased in mass? Explain.



$c(\text{dilute } H_2SO_4) = \frac{n}{V} = \frac{2.35 \times 10^{-3}}{0.0247}$
$= 0.0950 \text{ mol L}^{-1}$
$n(\text{dilute } H_2SO_4)$ *in* $1.00 \text{ L} = cV$
$= 0.0950 \times 1.00 = 0.0950 \text{ mol}$
$\therefore n(\text{conc. } H_2SO_4)$ *in* $20.0 \text{ mL}$
$= n(\text{dilute } H_2SO_4)$ *in* $1.00 \text{ L}$
$= 0.0950 \text{ mol}$

$c(\text{battery acid}) = \frac{n}{V} = \frac{0.0950}{0.0200}$
$= 4.75 \text{ mol L}^{-1}$

34. *At equivalence point the solution is neutral*
*i.e.* $[H^+] = 1.00 \times 10^{-7} \text{ mol L}^{-1}$
$V(\text{solution}) = 0.0249 + 0.0200 = 0.0449 \text{ L}$
$\therefore n(H^+)$ *at equivalence point* $= cV$
$= 1.00 \times 10^{-7} \times 0.0449 = 4.49 \times 10^{-9} \text{ mol}$
$n(H^+)$ *in 1 drop* $= cV = 0.100 \times 0.0001$
$= 1.00 \times 10^{-5} \text{ mol}$
$\therefore n(H^+)$ *in solution after 1 drop added*
$= 4.49 \times 10^{-9} + 1.00 \times 10^{-5} = 1.00 \times 10^{-5} \text{ mol}$

$c(H^+) = \frac{n}{V} = \frac{1.00 \times 10^{-5}}{0.0449}$

$= 2.23 \times 10^{-4} \text{ mol L}^{-1}$
*new* $pH = -\log(2.23 \times 10^{-4}) = 3.65$
$\therefore$ *change in* $pH = 7.00 - 3.65 = 3.35$

***For the Experts***

35.
(a) $Na_2CO_3 + 2HCl \rightarrow 2NaCl + H_2O + CO_2$
(b) $n(HCl) = cV = 0.136 \times 0.0226$
$= 3.07 \times 10^{-3} \text{ mol}$
(c) $n(Na_2CO_3)$ *in* $20 \text{ mL} = 1/2n(HCl)$
$= 1.54 \times 10^{-3} \text{ mol}$
(d) $n(Na_2CO_3)$ *in 500 mL of solution*

$= \frac{500}{20} \times 1.54 \times 10^{-3} = 3.84 \times 10^{-2} \text{ mol}$

$m(Na_2CO_3)$ *in 500 mL of solution*
$= nM = 3.84 \times 10^{-2} \times 105.99$
$= 4.07 \text{ g}$
$\therefore m(H_2O)$ *in* $Na_2CO_3.xH_2O$
$= m(Na_2CO_3.xH_2O) - m(Na_2CO_3)$
$= 10.96 - 4.07 = 6.89 \text{ g}$
(e) $n(H_2O)$ *in* $Na_2CO_3.xH_2O$

$= \frac{6.89}{18.016} = 0.382 \rightarrow \frac{n(H_2O)}{n(Na_2CO_3)} = \frac{0.382}{3.84 \times 10^{-2}} \text{ mol}$

*i.e.* $n(H_2O) = 10 \times n(Na_2CO_3)$
$\therefore$ *correct formula* $= Na_2CO_3.10H_2O$

## CHP 3: OXIDATION AND REDUCTION

### Chapter Questions

3.1
(a) $Mg(s) + Cl_2(g) \rightarrow MgCl_2(s)$
(b) $Mg$ (c) $Mg$ (d) $Cl_2$

3.2 *True: oxidant removes electrons from reductant, e.g.* $2Fe + O_2 \rightarrow 2FeO$
$O_2$ *removes electrons from Fe to form* $O^{2-}$ *and* $Fe^{2+}$.

3.3
(a) (+6) (-2) (b) (+6) (-2)
(c) (+5) (-2) (d) (-3) (+1) (-1)
(e) (+1) (+4) (-2) (f) (+2) (+4) (-2)

3.4
(a) *Redox* (b) *Not Redox*
(c) *Not Redox* (d) *Redox*

3.5
(a) *oxidation:* $Zn(s) \rightarrow Zn^{2+}(aq) + 2e^-$
*reduction:* $2H^+(aq) + 2e^- \rightarrow H_2(g)$
(b) *oxidation:* $Mg(s) \rightarrow Mg^{2+}(s) + 2e^-$
*reduction:* $S(s) + 2e^- \rightarrow S^{2-}(s)$

3.6
(a) $MnO_4^-(aq) + 8H^+(aq) + 5e^- \rightarrow Mn^{2+}(aq) + 4H_2O(l)$
(b) $H_2C_2O_4(aq) \rightarrow 2CO_2(g) + 2H^+(aq) + 2e^-$
(c) $CH_3CH_2OH(l) \rightarrow CH_3CHO(l) + 2H^+(aq) + 2e^-$

3.7
(a) $(MnO_4^-(aq) + 8H^+(aq) + 5e^- \rightarrow Mn^{2+}(aq) + 4H_2O(l)) \times 2$
$(H_2S(g) \rightarrow S(s) + 2H^+(aq) + 2e^-) \times 5$
$2MnO_4^-(aq) + 5H_2S(g) + 6H^+(aq) \rightarrow 2Mn^{2+}(aq) + 5S(s) + 8H_2O(l)$
(b) $ClO^- + 2H^+ + 2e^- \rightarrow Cl^- + H_2O$
$SO_2 + 2H_2O \rightarrow SO_4^{2-} + 4H^+ + 2e^-$
$ClO^-(aq) + SO_2(g) + H_2O(l) \rightarrow Cl^-(aq) + SO_4^{2-}(aq) + 2H^+(aq)$

3.8
(a) $5Fe^{2+}(aq) + MnO_4^-(aq) + 8H^+(aq) \rightarrow 5Fe^{3+}(aq) + Mn^{2+}(aq) + 4H_2O(l)$
(b) $Sn^{2+}(aq) + 2H^+(aq) + OCl^-(aq) \rightarrow Cl^-(aq) + Sn^{4+}(aq) + H_2O(l)$

3.9
$$\begin{array}{c} 2ClO_3^- + 12H^+ + 10e^- \rightarrow Cl_2 + 6H_2O \\ (ClO_3^- + H_2O \rightarrow ClO_4^- + 2H^+ + 2e^-) \times 5 \\ \hline 7ClO_3^-(aq) + 2H^+(aq) \rightarrow Cl_2(g) + 5ClO_4^-(aq) + H_2O(l) \end{array}$$

3.10
$$\begin{array}{c} 4OH^- + Cl_2 \rightarrow 2OCl^- + 2e^- \\ Cl_2 + 2e^- \rightarrow 2Cl^- \\ \hline 4OH^- + 2Cl_2 \rightarrow 2Cl^-(aq) + 2OCl^-(aq) + 2H_2O(l) \end{array}$$

3.11

$(Cu \rightarrow Cu^{2+} + 2e^-) \times 3$ $\quad -0.34$ V

$(NO_3^- + 4H^+ + 3e^- \rightarrow NO + 2H_2O) \times 2$ $\quad +0.96$ V

$2NO_3^- + 3Cu + 8H^+ \rightarrow 3Cu^{2+} + 2NO + 4H_2O$ $\quad +0.62$ V

3.12

$Fe \rightarrow Fe^{2+} + 2e^-$ $\quad E° = +0.44$ V

$Cu^{2+} + 2e^- \rightarrow Cu$ $\quad \underline{E° = +0.34 \text{ V}}$

$Cu^{2+} + Fe \rightarrow Fe^{2+} + Cu$ $\quad E°_{TOT} = +0.78$ V

*Not feasible → the steel tank will react. Alloying alters likely reactions.*

3.13

(a) $Fe \rightarrow Fe^{2+} + 2e^-$ $\quad +0.44$ V

$Zn^{2+} + 2e^- \rightarrow Zn$ $\quad \underline{-0.76 \text{ V}}$

*No reaction* $\quad -0.32$ V

(b) $Cu \rightarrow Cu^{2+} + 2e^-$ $\quad +0.44$ V

$NO_3^- + 4H+ + 3e^- \rightarrow NO + 2H_2O$ $\quad \underline{+0.96 \text{ V}}$

$+1.40$ V

*Reaction will occur.*

(c) $Br_2 + 2e^- \rightarrow 2Br^-(aq)$ $\quad +1.08$ V

$2Cl^- \rightarrow Cl_2 + 2e^-$ $\quad \underline{-1.36 \text{ V}}$

*Reaction will not occur.* $\quad -0.32$ V

(d) $Cl_2 + 2e^- \rightarrow 2Cl^-$ $\quad +1.36$ V

$H_2S \rightarrow S + 2H^+ + 2e^-$ $\quad \underline{-0.17 \text{ V}}$

*Reaction does occur.* $\quad +1.25$ V

(e) $2I^- \rightarrow I_2 + 2e^-$ $\quad -0.54$ V

$MnO_4^- + 8H^+ + 5e^- \rightarrow Mn^{2+} + 4H_2O$ $\quad \underline{+1.51 \text{ V}}$

*Reaction does occur.* $\quad +0.97$ V

(f) $Ca \rightarrow Ca^{2+} + 2e^-$ $\quad +2.87$ V

$2H_2O + 2e^- \rightarrow H_2 + 2OH^-$ $\quad \underline{-0.83 \text{ V}}$

*Reaction does occur.* $\quad +2.04$ V

(g) $MnO_4^- + 8H^+ + 5e^- \rightarrow Mn^{2+} + 4H_2O$ $\quad +1.51$ V

$Fe^{2+} \rightarrow Fe^{3+} + 2e^-$ $\quad \underline{-0.77 \text{ V}}$

*Reaction does occur.* $\quad +0.74$ V

3.14

| Common Oxidising Agents | Name | Reduction Half Equation | E° |
|---|---|---|---|
| $Cl_2$ | chlorine | $Cl_2 + 2e^- \rightarrow 2Cl^-$ | +1.36V |
| $MnO_4^-$ | permanganate ion | $MnO_4^- + 8H^+ + 5e^- \rightarrow Mn^{2+} + 4H_2O$ | +1.51V |
| $Cr_2O_7^{2-}$ | dichromate ion | $Cr_2O_7^{2-} + 14H^+ + 6e^- \rightarrow 2Cr^{3+} + 7H_2O$ | +1.36V |
| $ClO^-$ | hypochlorite ion | $ClO^- + H_2O + 2e^- \rightarrow Cl^- + 2OH^-$ | +0.90V |
| $H^+$ | hydrogen ion | $2H^+ + 2e^- \rightarrow H_2$ | 0.00V |

3.15

| Common Reducing Agents | Name | Reduction Half Equation | E° |
|---|---|---|---|
| Mg | magnesium | $Mg \rightarrow Mg^{2+} + 2e-$ | +2.36 |
| Zn | zinc | $Zn \rightarrow Zn^{2+} + 2e^-$ | +0.76V |
| $C_2O_4^{2-}$ | oxalate ion | $C_2O_4^{2-} \rightarrow 2CO_2 + 2e^-$ | +0.43V |
| $H_2$ | hydrogen gas | $H_2 \rightarrow 2H^+ + 2e^-$ | 0.00V |
| $Fe^{2+}$ | iron (II) ion | $Fe^{2+} \rightarrow Fe^{3+} + e^-$ | -0.77V |

3.16

(a) $Zn \rightarrow Zn^{2+} + 2e^-$

(b) (i) Zn strip will be blackened, dull salmon pink solid growing on it.
(ii) Solution loses blue colour.

(c) (i) $Cu^{2+}$
(ii) $Zn$

3.17

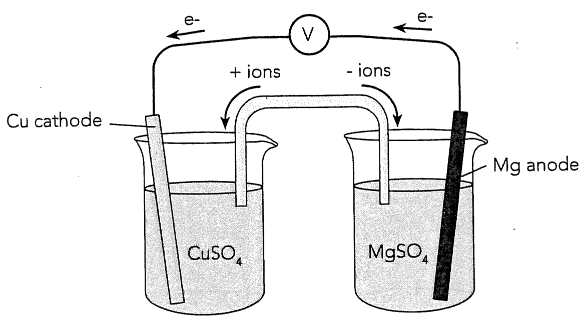
(a) To allow a flow of ions to complete the circuit.

(b) (i) $Zn^{2+}$, $NH_4^+$ ions move away from $Zn/Zn^{2+}$ half cell.
(ii) $SO_4^{2-}$, $NO_3^-$ ions move away from $Cu/Cu^{2+}$ half cell.

(c)
- slowly dissolves
- slowly gains mass
- no change
- loses blue colour

3.18

(a)




(b) $Mg \rightarrow Mg^{2+} + 2e^-$
$Cu^{2+} + 2e^- \rightarrow Cu$
$Mg(s) + Cu^{2+}(aq) \rightarrow Mg^{2+}(aq) + Cu(s)$

(c) 2.71 V

(d) (i) Chemical equilibrium has been reached.
(ii) Reactants are consumed.

*Dry Cell – Overall Reaction*

$2MnO_2(s) + Zn(s) + 2H^+(aq) \rightarrow Mn_2O_3(s) + Zn^{2+}(aq) + H_2O(l)$

3.19

(a) $NH_4^+(aq) \rightarrow NH_3(aq) + H^+(aq)$

(b) Once it has been discharged it cannot be recharged.

(c) Advantages: cheap, transportable.
Disadvantages: relatively short life, cheap casing can dissolve, hence allowing leakage.

3.20

$Zn \rightarrow Zn^{2+} + 2e^-$
$Ag_2O + H_2O + 2e^- \rightarrow 2Ag(s) + 2OH^-(aq)$

*Lead-Acid Accumulator – Overall Reaction for Discharge*

$Pb + PbO_2 + 2SO_4^{2-} + 4H^+ \rightarrow 2PbSO_4 + 2H_2O$

3.21

$2PbSO_4(s) + 2H_2O(l) \rightarrow Pb(s) + PbO_2(s) + 4H^+(aq) + 2SO_4^{2-}(aq)$

3.22

(i) Electrodes are same substance – no reaction to produce transfer of $e^-$.

(ii) Density of electrolyte decreases.

*Fuel Cell – Overall Reaction*

$2H_2(g) + O_2(g) \rightarrow 2H_2O(l)$

3.23

(a) 1.23 V

(b) $H_2 + 2OH^- \rightarrow 2H_2O + 2e^-$
$O_2 + 2H_2O + 4e^- \rightarrow 4OH^-$

(c) $2H_2 + O_2 \rightarrow 2H_2O$

(d) Same reaction but hydrogen combustion is explosive.

3.24

$CH_4 + 2H_2O \rightarrow CO_2 + 8H^+ + 8e^-$
$(O_2 + 2H_2O + 4e^- \rightarrow 2H_2O) \times 2$
$CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$

3.25

(a) Possible cathode reactions

| | |
|---|---|
| $Ni^{2+}(aq) + 2e^- \rightarrow Ni(s)$ | -0.26 V |
| $2H_2O(l) + 2e^- \rightarrow H_2(g) + 2OH^-(aq)$ | -0.83 V |

Possible anode reactions

| | |
|---|---|
| $Cu(s) \rightarrow Cu^{2+}(aq) + 2e^-$ | -0.34 V |
| $2Cl^-(aq) \rightarrow Cl_2(g) + 2e^-$ | -1.36 V |
| $2H_2O(l) \rightarrow O_2(g) + 4H^+(aq) + 4e^-$ | -1.23 V |

Overall reaction
$Ni^{2+}(aq) + Cu(s) \rightarrow Ni(s) + Cu^{2+}(aq)$

(b) Concentration of $Ni^{+2}$ ions in solution will decrease while that of $Cu^{+2}$ ions will increase. As this occurs $Cu^{+2}$ ions will be preferentially reduced at the cathode (more positive $E^\circ$ than $Ni^{+2}$ or $H_2O$)

3.26

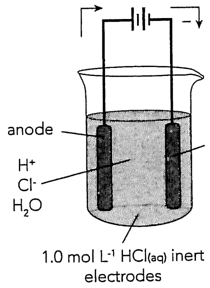
(a) Aqueous NaCl (1.0 mol $L^{-1}$)
anode: $2H_2O(l) \rightarrow O_2(g) + 4H^+(aq) + 4e^-$
cathode: $2H_2O(l) + 2e^- \rightarrow H_2(g) + 2OH^-(aq)$
Products are $O_2$ and $H_2$.

(b) Aqueous NaCl (concentrated)
anode: $2Cl^-(aq) \rightarrow Cl_2(g) + 2e^-$
cathode: $H_2O(l) + 2e^- \rightarrow H_2(g) + 2OH^-(aq)$

Products are $Cl_2$, $H_2$ and NaOH solution.

3.27

(a)




(b) anode:

$2Cl^-(aq) \rightarrow Cl_2(g) + 2e^-$ -1.36 V

$2H_2O(l) \rightarrow O_2(g) + 4H^+(aq) + 4e^-$ -1.23 V

cathode:

$2H^+(aq) + 2e^-$ 0.00 V

$2H_2O(l) + 2e^- \rightarrow H_2(g) + 2OH^+(aq)$ -0.83 V

Overall reaction

$2H_2O(l) \rightarrow O_2(g) + 2H_2(g)$

3.28

(a) Molten NaCl

anode: $2Cl^-(l) \rightarrow Cl_2(g) + 2e^-$

cathode: $Na^+(l) + e^- \rightarrow Na(l)$

Products are $Cl_2$ and Na.

(a) Molten $PbI_2$

anode: $2I^-(l) \rightarrow I_2(l)$

cathode: $Pb^{2+}(l) + 2e- \rightarrow Pb$

Products are $I_2$ and Pb.

3.29

anode: $Ag(s) \rightarrow Ag^+(aq) + e-$

cathode: $Ag^+(aq) + e- \rightarrow Ag(s)$

Overall reaction

$Ag(s)$ anode $\rightarrow Ag(s)$ cathode

3.30

The equilibrium reaction keeps the concentration of $Ag^+$ ions in solution low but fairly constant. As $Ag^+$ ions are deposited at the cathode more of the $Ag(CN)_2^-$ dissociates to maintain equilibrium. This keeps the $Ag^+$ ions in solution fairly constant.

3.31

(a) anode: $Cu(s) \rightarrow Cu^{2+}(aq) + 2e^-$

cathode: $Cu^{2+}(aq) + 2e^- \rightarrow Cu(s)$

Overall reaction

(b) (i) E° for their oxidation is much higher than Cu.

(ii) These valuable metals are recovered from the anode mud for profit.

(c) (i) Only a small voltage is used which is not high enough to reduce these ions.

(ii) The $Pb^{+2}$ ions react with the sulphate ions and are recovered as lead sulphate residue. The electrolyte solution is periodically replaced as the $Ni^{+2}$ ion concentration increases. The $Ni^{+2}$ ions are removed as nickel sulphate solution.

3.32 Au, Ag, Cu, Pb, Fe, Al, Mg, Na

## 3. Review Questions

1. (a) +2 (b) +6 (c) -1 (d) +3
   (e) +1 (f) +3 (g) +2
2. (a) H (b) Na (c) Cl
   (d) nil (e) C
3. (a) O (b) N (c) O (d) nil

4.

| Oxidising Agents | Reducing Agents |
|---|---|
| oxygen gas, chlorine gas, acidified potassium permanganate, concentrated sulfuric acid, potassium dichromate (acidified), concentrated nitric acid, hydronium ion | zinc, hydrogen gas, carbon, iron (II) ions oxalic acid |

5.

| Equation | Colour of Reactants | Colour of Products |
|---|---|---|
| $Br_2(aq) + 2I^-(aq) \rightarrow 2Br^-(aq) + I^2(aq)$ | straw brown | red brown |
| $2Br^-(aq) + Cl_2(aq) \rightarrow 2Cl^-(aq) + Br_2(aq)$ | colourless | straw brown |
| $Cl_2(aq) + 2I^-(aq) \rightarrow 2Cl^-(aq) + I^2(aq)$ | colourless | red brown |
| $2Ag^+(aq) + Cu(s) \rightarrow 2Ag(s) + Cu^{2+}(aq)$ | colourless solution, reddish metal | light blue solution, silvery black crystals |
| $Mg(s) + Cu^{2+}(aq) \rightarrow Mg^{2+}(aq) + Cu(s)$ | blue solution | soln loses colour, brown black crystals grow |

6.

(a) $2S_2O_3^{2-}(aq) + I_2(s) \rightarrow S_4O_6^{2-}(aq) + 2I^-(aq)$

(b) $2Br^-(aq) + SO_4^{2-}(aq) + 4H^+(aq) \rightarrow Br_2(l) + SO_2(g) + 2H_2O(l)$

(c) $3Ag_2S(s) + 2NO_3^-(aq) + 8H^+(aq) \rightarrow 6Ag^+(aq) + 3S(s) + 2NO(g) + 4H_2O(l)$

(d) $2CrO_4^{2-}(aq) + 6Cl^-(aq) + 16H^+(aq) \rightarrow 2Cr^{3+}(aq) + 3Cl_2(g) + 8H_2O(l)$

(e) $MnCl_2(s) + Br_2(aq) + 4OH^-(aq) \rightarrow MnO_2(s) + 2Br^-(aq) + 2Cl^-(aq) + 2H_2O(l)$

7.

(a) $Ag(s) + 2H^+(aq) + NO_3^-(aq) \rightarrow Ag^+(aq) + NO_2(g) + H_2O(l)$

(b) $S(s) + Cr_2O_7^{2-}(aq) \rightarrow SO_4^{2-}(aq) + Cr_2O_3(s)$

(c) $3H_2S(s) + 2NO_3^-(aq) + 2H^+(aq) \rightarrow 3S(s) + 2NO(g) + 4H_2O(l)$

(d) $2ClO^-(aq) + 4H^+(aq) + 2Cl^-(aq) \rightarrow 2Cl_2(g) + 2H_2O(l)$

(e) $5Fe^{2+}(aq) + MnO_4^-(aq) + 8H^+(aq) \rightarrow 5Fe^{3+}(aq) + Mn^{2+}(aq) + 4H_2O(l)$

(f) $Cr_2O_7^{2-}(aq) + 14H^+(aq) + 6I^-(aq) \rightarrow 2Cr^{3+}(aq) + 3I_2(aq) + 7H_2O(l)$

(g) $2MnO_4^-(aq) + 5H_2O_2(aq) + 6H^+(aq) \rightarrow 2Mn^{2+}(aq) + 5O_2(g) + 8H_2O(l)$

8. When an element is simultaneously oxidised and reduced, disproportionation is said to occur, e.g.

$H_2O_2(aq) + 2H^+(aq) + 2e^- \rightarrow 2H_2O_2(l)$
$H_2O_2(aq) \rightarrow O_2(g) + 2H^+(aq) + 2e^-$

$2H_2O_2 \rightarrow O_2 + 2H_2O$
(-1) (0) (-2)

*i.e. the oxygen in hydrogen peroxide has been oxidised to form $O_2$ and reduced in forming $H_2O$.*

9. *May be used as bleaches or for sterilising swimming pool water and drinking water*
10. *Chlorine, gold, silver, hydrogen, lead, calcium.*
11. *$Br^-$ and $I^-$.*
12. *$K^+$, $Cr^{3+}$, $Br_2$, $Cr_2O_7^{2-}$, $Au^{3+}$, $MnO_4^-$, $H_2O_2$.*
13.
(a) *All oxidising agents above $Sn^{4+} + 2e^- \rightarrow Sn^{2+}$ in a table of a standard reduction potentials.*
(b) *All reducing agents below $Fe^{2+} + Fe^{3+} \rightarrow e^-$ in a table of standard reduction potentials.*
(c) *Acidified $MnO_4^-$ (only just), HClO, $H_2O_2$ or $F_2$ (by use of a table of standard reduction potentials).*
(d) *Acidified $NO_3^-$, $Hg^{2+}$, $Br_2$.*
14. *A galvanic cell may be established and the Al oxidises causing the rivets to break.*
15.
(a) *The standard electrode potential is the potential acquired if a chemical (eg. Al(s)) is immersed in a solution of its ions (e.g. $Al^{3+}$ ions) of concentration 1.00 mol $L^{-1}$ at 25°C. E° values are a comparison between the standard reduction potential of a substance and the $H_2/H^+$ reference cell.*
(b) *No – if reduction is occurring, oxidation must occur simultaneously.*
(c) *Mg, Na.*

16.
(a) *No reaction.*
(b) $(Cu^+ \rightarrow Cu^{2+} + e^-) \times 2$
$I_2 + 2e^- \rightarrow 2I^-$

$2Cu^+(aq) + I_2(aq) \rightarrow 2Cu^{2+}(aq) + 2I^-(aq)$

(c) $(MnO_4^- + 8H^+ + 5e^- \rightarrow Mn^{2+} + 4H_2O) \times 2$
$(H_2O_2 \rightarrow O_2 + 2H^+ + 2e^-) \times 5$

$2MnO_4^-(aq) + 5H_2O_2 + 6H^+(aq) \rightarrow 2Mn^{2+}(aq) + 5O_2(g) + 8H_2O(l)$

(d) $Cu \rightarrow Cu^{2+} + 2e^-$
$(NO_3^- + 2H^+ + e^- \rightarrow NO_2(g) + H_2O(l)) \times 2$

$Cu(s) + 2NO_3^-(aq) + 4H^+(aq) \rightarrow Cu^{2+}(aq) + 2NO_2(g) + 2H_2O(l)$

(e) $(Na \rightarrow Na^+ + e^-) \times 2$
$2H_2O + 2e^- \rightarrow H_2 + 2OH^-$

$2Na(s) + 2H_2O(l) \rightarrow 2Na^+(aq) + H_2(g) + 2OH^-(aq)$

(f) $(MnO_4^- + 8H^+ + 5e^- \rightarrow Mn^{2+} + 4H_2O) \times 2$
$(2Cl^- + Cl_2 + 2e^-) \times 5$
$2MnO_4^-(aq) + 10Cl^-(aq) + 16H^+(aq) \rightarrow 2Mn^{2+}(aq) + 5Cl_2(g) + 8H_2O(l)$

(g) *No reaction.*
(h) *Boiling removes $O_2$; iron will not corrode.*
$Fe + Fe^{2+} \rightarrow 2e^-$ +0.44 V
$2H_2O + 2e^- \rightarrow H_2 + 2OH^-$ −0.41 V
*+0.03 V potential, will occur very slowly.*

17. *They do initially – but once an oxide layer is formed it will protect the metal beneath from further oxidation. This oxide layer is a coherent coating.*

18. *(a) oxidation (b) reduction (c) negative (d) positive*

19.
(a) *From Mg → Pb* (b) *electrons*
(c) *ions ($Mg^{2+}$, $Pb^{2+}$, $NO_3^-$)* (d) *$KNO_3$*
(e) $Mg(s) \rightarrow Mg^{2+}(aq) + 2e^-$
(f) $Pb^{2+}(aq) + 2e^- \rightarrow Pb(s)$ (g) +2.24 V
(h) *To complete the circuit by allowing the movement of ions.*
20.
(a) *Yes*
(b) *Yes, but no current would flow in the external circuit, reading on meter = 0.*

21.
(a) Anode: $Zn(s) \rightarrow Zn^{2+} + 2e^-$ (0.76)
Cathode: $Cu^{2+} + 2e^- \rightarrow Cu(s)$ (0.34)
E° = 1.10 V
(b) Anode: $Fe(s) \rightarrow Fe^{2+} + 2e^-$ (0.44)
Cathode: $Ag^+ + e^- \rightarrow Ag(s)$ (0.80)
E° = 1.24 V
(c) Anode: $Al(s) \rightarrow Al^{3+} + 3e^-$ (1.68)
Cathode: $I_2(aq) + 2e^- \rightarrow 2I^-$ (0.54)
E° = 2.22 V
(d) Anode: $2Br \rightarrow Br_2 + 2e^-$ (−1.08)
Cathode: $F_2(g) + 2e^- \rightarrow 2F^-$ (2.89)
E° = 1.81 V
(e) Anode: $H_2O_2 \rightarrow O_2 + 2H^+ + 2e^-$ (−0.70)
Cathode: $MnO_4^- + 8H^+ + 5e^- \rightarrow Mn^{2+} + 4H_2O$ (1.51)

E° = 0.81 V

22.

| Electrochemical Cells | Electrolytic Cells |
|---|---|
| Spontaneous chemical reaction | Forced chemical reaction |
| oxidation occurs at the anode | oxidation occurs at the anode |
| anode is negative | anode is positive |
| electrical energy is created by a chemical change | a chemical change is caused by electrical energy |

23.
(a) $Na^+(l) + e^- \rightarrow Na^+(l)$
$2Cl^-(l) \rightarrow Cl(g) + 2e^-$ *min volts = 4.07 V*
(b) $Zn^{2+}(l) + 2e^- \rightarrow Zn(l)$
$2Br^-(l) \rightarrow Br_2(g) + 2e^-$ *min volts = 1.84 V*
(c) $Pb^{2+}(l) + 2e^- \rightarrow Zn(l)$
$2I^-(l) \rightarrow I_2(g) + 2e^-$ *min volts = 0.67 V*

(d) $K^+(l) + e^- \rightarrow K(l)$
$2F^-(l) \rightarrow F_2(g) + 2e^-$ min volts = 5.83 V

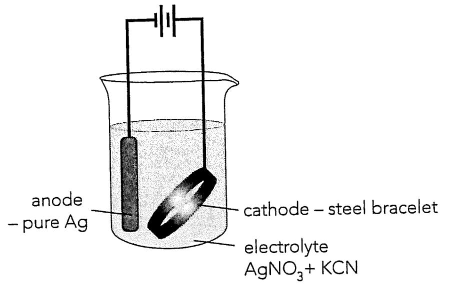
24.
(a) $Na^+$, $Cl^-$, $H_2O$ (b) $Cl^-$ and $H_2O$
(c) $H_2O$
(d) $Na^+$ and $H_2O$; $H_2O$ most likely
(e) $(2H_2O) + 2e^- \rightarrow H_2 + 2OH^-) \times 2$
$E° = -.083 V$ (-0.41 V at pH=7)
$2H_2O \rightarrow O_2 + 4H^+ + 4e^-$
$\underline{E° = -1.23 V\ (-0.82\ V\ at\ pH=7)}$
$2H_2O(l) \rightarrow O_2(g) + 2H_2(g)$

(f) 2.06 V minimum (-1.23 V at pH=7)

25. (c) $H_2O$ is oxidised (d) $H_2O$ is reduced

26.
(a) oxidation: $2I^-(aq) \rightarrow I_2(s) + 2e^-$ $E° = -0.54$ V
reduction: $Ni^{2+}(aq) + 2e^- \rightarrow Ni(s)$ $E° = -0.26$ V
$2I^-(aq) + Ni^{2+}(aq) \rightarrow I_2(s) + Ni(s)$ $E° = -0.80$ V
(b) oxidation: $Pb(s) \rightarrow Pb^{2+} + 2e^-$ $E° = +0.13$ V
reduction: $Pb^{2+} + 2e^- \rightarrow Pb(s)$ $E° = -0.13$ V
$Pb(s) \rightarrow Pb(s)$ $E° = 0.0$ V
Lead is effectively being transferred from the anode to the cathode. Very small positive voltage needed.
(c) oxidation:
$2H_2O(l) \rightarrow O_2(g) + 4H^+(aq) + 4e^-$ $E° = -1.23$ V
$(2H^+(aq) + 2e^- \rightarrow H_2) \times 2$ $E° = 0.00$ V
$2H_2O(l) \rightarrow O_2(g) + 2H_2(g)$
(d) anode:
$Cu(s) \rightarrow Cu^{2+}(aq) + 2e^-$ $E° = -0.34$ V
$2H_2O(l) + 2e^- \rightarrow H_2(g) + 2OH^-(aq)$ $E° = -0.83$ V
$Cu^{2+}(aq) + H_2(g) + 2OH^-(aq)$ $E° = -1.17$ V

27. Make the steel the cathode:
$Ag^+(aq) + e^- \rightarrow Ag(s)$
The steel will be coated with Ag.




28. $H_2SO_4$: $2H_2O(l) \rightarrow 2H_2(g) + O_2(g)$
NaOH: $2H_2O(l) \rightarrow 2H_2(g) + O_2(g)$

29.
- reduced
- cathode, $O_2 + 2H_2O + 4e^- \rightarrow 4OH^-$
- iron (II) hydroxide
- $Fe_2O_3.H_2O$ or FeO(OH)

30. The reduction process is:
$O_2(g) + 2H_2O(l) + 4e^- \rightarrow 4OH^-$
By having an alkaline environment the reverse reaction is favoured and reduction process is less likely to occur.

31.
(a) It coats the iron and stops contact with $O_2$ and $H_2O$, hence stopping the iron from corroding.
(b) $Fe \rightarrow Fe^{2+} + 2e^-$, $E° = +0.44$ V
$Sn(s) \rightarrow Sn^{2+} + 2e^-$, $E° = +0.14$ V
$O_2 + 2H_2O + 4e^- \rightarrow 4OH^-$, $E° = +0.82$ V
Once the Fe is exposed to $O_2$ and $H_2O$ it will oxidise in preference to the Sn; i.e. the Fe will protect the Sn from oxidation.
(c) $Fe \rightarrow Fe^{2+} + 2e^-$, $E° = +0.44$ V
$Zn \rightarrow Zn^{2+} + 2e^-$, $E° = +0.76$ V
The Zn will oxidise before the Fe, protecting the Fe.

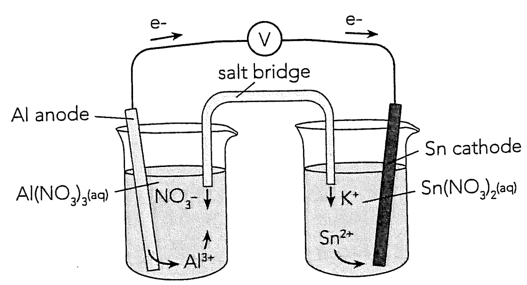
32. The Mg (or Zn) has a higher oxidation potential than the Fe and so will be oxidised before the Fe.

33. Add coating of oil: prevents $O_2$ and $H_2O$ coming in contact with the Fe and so reduces the likelihood of corrosion.
Clean drain holes: stops $H_2O$ build up which could promote rusting.

34.
(a) Oxidation occurs at the anode; by making the pipe the cathode the likelihood of it corroding is reduced.
(b) The metals that are connected as the anode will oxidise and need replacement at some stage.

35. By the oxidation of C and $CO_2$.
$C(s) + O_2(g) \rightarrow CO_2(g)$
$C(s) + CO_2(g) \rightarrow 2CO(g)$
(b) $Fe_2O_3(s) + 3CO_2(g) \rightarrow 2Fe(l) + 3CO_2(g)$

36. (a), (d)




(b) Anode: $Al(s) \rightarrow Al^{3+}(aq) + 3e^-$
Cathode: $Sn^{2+}(aq) + 2e^- \rightarrow Sn(s)$
(c) $E° = +1.54$ V
(d) See diagram at (a).
(e)
- The concentration of the reactant ions in solution will have reduced.
- The electrode surfaces become pitted or coated and reduce reaction rate.

(f) The Sn electrode would have gained mass as the $Sn^{2+}$ ions form Sn(s) on its surface.